

Introduction

Dean Lusher, Johan Koskinen, and Garry Robins

Exponential random graph models¹ (ERGMs) are a class of statistical model for social networks. They account for the presence (and absence) of network ties, and so provide a model for network structure. An ERGM models a given network in terms of small local tie-based structures, such as reciprocated ties and triangles. A social network can be thought of as being built of these local patterns of ties, called “network configurations,” which correspond to the parameters in the model. Moreover, these configurations can be considered to arise from local social processes, whereby actors in the network form connections in response to other ties in their social environment. ERGMs are a principled statistical approach to modeling social networks. They are theory driven in that their use requires the researcher to consider the complex, intersecting, and, indeed, potentially competing theoretical reasons why the social ties in the observed network have arisen. For instance, does a given network structure occur due to processes of homophily, reciprocity, transitivity, or a combination of these? By including such parameters together in the one model, a researcher can test these effects one against the other, and so infer the social processes that have built the network. Being a statistical model, an ERGM permits inferences about whether, in our network of interest, there are significantly more (or less) reciprocated ties, or triangles (for instance), than we would expect.

ERGMs are fast becoming recognized as one of the central approaches in analyzing social networks. In this short introductory chapter, we describe the intent of this book, how it is structured, how it may be read, the software resources available, and the knowledge that we expect readers to have before following our text.

¹ Throughout this book, we use this now-established convenient term instead of the more correct (but cumbersome) “exponential family random graph models.”

1.1 Intent of This Book

In this book, we aim to introduce ERGMs in a way that takes the reader from the basic theoretical assumptions underlying the models, through the technical specifications and statistical detail, to applied examples illustrating how various substantive research questions may be investigated and tested empirically. The goals of this book are twofold: to describe ERGMs and to demonstrate how recent methodological developments allow us to address social network research questions in new and powerful ways. A specific target audience is the growing number of social scientists who are interested in statistical models for social networks and network-based social processes. We aim to provide an intuitive understanding of these models for those readers who may be unfamiliar with ERGMs. We also focus on delivering sufficient technical detail for those with a social network methodological background and who are interested in a deeper understanding of the modeling and estimation. We hope to lay bare the value of a statistical modeling approach in answering core questions about interactive social processes.

1.2 Software and Data

The book includes references to data and software that can be downloaded so that readers can reproduce some of the applications in a “hands-on” fashion. Some parts of this book make explicit use of the PNet suite of programs for ERGMs (Wang, Robins, & Pattison, 2009), and we tend to use PNet terminology for parameters and the like, but the exposition is in no way contingent on the software used (as long as the software does deal with ERGMs, naturally). The estimation algorithms used in PNet and in the R package *statnet* (Handcock et al., 2003; Handcock et al., 2008) are both described in Chapter 12, as is the principle of *Bergm* (Caimo & Friel, 2011). *SIENA* 3 can be used for fitting ERGMs to cross-sectional data (indeed, the main parts of PNet draw heavily on *SIENA*) and longitudinal models can be analyzed in *RSIENA* (or *SIENA* 4). Models for multiple networks, longitudinal networks, and bipartite networks can be estimated from the PNet suite of programs, *XPNet*, *LPNet*, and *BPNet*, respectively. The autologistic actor attribute models require *iPNet*. PNet and the example data used in this book can be downloaded to help you work through the chapters in a practical, hands-on fashion (<http://www.sna.unimelb.edu.au>). The package *statnet* is an ERGM estimation program in the R environment and can be installed from CRAN in standard R fashion. A useful introduction to the *statnet* package is given in a special issue of *Journal of Statistical*

Software (Hunter et al., 2008), and the worked-through examples of Goodreau et al. (2008) are particularly helpful.

1.3 Structure of the Book

This book is divided into four main sections that broadly map onto the rationale and theory, method, and application of ERGMs, with a fourth overview section discussing future directions. We consider that theory, method, and empirical work are fundamentally intertwined and interdependent. Thus, you will note that the first section, despite its flavor of conceptual discussion, still addresses methodological issues and provides empirical examples, whereas theoretical concepts permeate the methods section.

1.3.1 Section I: Rationale

Section I of this book provides an intuitive introduction to ERGMs by connecting to various aspects of social network (and social science) theory, the thinking behind the methods, and the empirical examination of research questions. [Chapter 2](#) provides a very general initial description of ERGMs, including some broad definitions and a discussion of some central elements of the ERGM approach. [Chapter 3](#) introduces some important network ideas about the formation of social ties and social structure more generally, and explains how these relate to an ERGM analysis. [Chapter 4](#) introduces some central methodological details of ERGMs in intuitive terms for readers unfamiliar with the models, with more technical details delayed until Section II on methods. Finally, in this section, [Chapter 5](#) provides an early and simple example of applying an ERGM to network data (an example we explore in greater detail in [Chapter 13](#)). In this way, we illustrate the type of inferences and interpretations that can be made about network structure using ERGMs. Our aims in this first section are not only to describe theoretical issues important to ERGMs but also to build intuitions for those coming to this modeling approach for the first time, before they encounter the more technical detail in Section II.

1.3.2 Section II: Methods

Section II of this book presents the basic statistical framework of ERGMs. Throughout the book, the case of a single observation on a (binary, unipartite) network is taken as a standard point of reference, and the technical details of ERGMs are introduced in the context of such a case

in [Chapter 6](#). [Chapter 7](#) seeks to give a deeper understanding of ERGMs by way of the notion of the dependence graph. From here, a range of more specific ERGMs are presented, beginning with models that include social selection, dyadic covariate, and geospatial effects in [Chapter 8](#). [Chapter 9](#) covers autologistic actor attribute models (ALAAMs), which are a version of social influence models, for the attributes of the actors. The next chapters describe models for multiple networks and bipartite networks ([Chapter 10](#)) as well as longitudinal network data ([Chapter 11](#)). The section concludes with descriptions of simulation, estimation, and goodness-of-fit procedures ([Chapter 12](#)), together with some illustrations ([Chapter 13](#)).

1.3.3 Section III: Applications

Section III of this book demonstrates how ERGMs can be applied empirically to answer social network research questions. In this section, we present an application for each of the different types of ERGMs presented previously. The chapters illustrate to the reader the range of possibilities, issues, and general frame of reference a researcher engages in when using ERGMs. The chapters are presented as separate self-contained case studies, but together they cover a set of network theoretical issues and features of the modeling framework.

In these chapters, we seek to show why and how the models can be used to answer novel and theoretically important questions, formulated in a relational framework. The range of issues covered gives some indication of the breadth of approach that ERGMs offer. For instance, Lusher and Robins ([Chapter 14](#)) note how the individual perceptions of attitudes held by others in the network have an independent association on the formation of network ties in two different contexts. They draw on theory from social psychology in their exposition. Lomi and Pallotti ([Chapter 15](#)) examine various path closure effects and, ultimately, show the importance of structural equivalence-type patterns in the transfer of patients among hospitals. Zhao and Rank ([Chapter 16](#)) examine how multiple types of relations within the one organization relate to each other. Drawing on various leadership theories, Kalish and Luria ([Chapter 17](#)) construe leadership as a relational phenomenon and show how leadership networks differ from many other types of social networks. Daraganova and Pattison ([Chapter 18](#)) test competing hypotheses about the impact of spatial factors and social ties on unemployment. Igarashi ([Chapter 19](#)) demonstrates the importance of homophily in a longitudinal analysis of networks involving different methods of communication. Using bipartite network models, Harrigan and Bond ([Chapter 20](#)) show how different forms of capital lead to different types of linking behavior. In the final applications chapter, Quintane

(Chapter 21) examines differences in self-reported ties and observed relational behaviors.

1.3.4 Section IV: Future

There is ongoing methodological development: this book is by no means the last word on ERGMs. To conclude this book, Pattison and Snijders look to future directions for ERGM research and methods.

1.4 How To Read This Book

We suggest that a cover-to-cover reading may not be the best way to approach the book for all readers. Readers with less statistical knowledge might begin with all four chapters in Section I: Rationale, which provides the general basis of ERGMs. Following this, the introductory methods chapter on the fundamentals of ERGMs (Chapter 6) is important, although it could be tackled later if need be. At the very least, the social selection models (Chapter 8) and the illustrations chapter (Chapter 13) in Section II: Methods should be read before moving to relevant chapters of Section III: Applications, and Section IV: Future.

Some readers, however, may want to understand ERGMs from the point of view of the statistical model. Thus, they may want to jump straight to the methods section, especially Chapters 6 and 7, and then work their way through Chapter 12 before returning to the conceptual issues of Section I.

For some readers whose interests are principally directed to fitting models empirically, the chapter on dependency (Chapter 7), which details important conceptual issues with regard to ERGMs, may be skipped at the first or first few readings. For those interested in the theoretical underpinning of the statistical model, however, this chapter is important.

For readers with some familiarity with the ERGM framework and an interest in a specific type of model, it is probably good to have skimmed through Section I first (and perhaps the first chapter of Section II). From there, the reader may go on to the applications chapter of choice, and then work him- or herself back to the relevant chapters in Section II, for clarification of details. You may, for example, have a particular interest in social influence-type models, in which case you might want to start with Chapter 18 and refer back to Chapter 9 in Section II, when necessary.

If you intend to fit your own ERGMs, it is necessary to understand how the models are fitted, whether as introduced in Chapters 4 and 5 of Section I and then Chapter 13, or more in-depth as in Chapter 12 (simulation, estimation, and goodness of fit). Specific applications chapters in Section III may also prove quite useful as a guide. Finally, once you have

fitted a model, or several, you may want to delve deeper into the details of Section II.

1.5 Assumed Knowledge of Social Network Analysis

We assume basic familiarity with social network analysis and that the reader understand the general concepts and terminology used in network and graph theory. For this purpose, we refer the reader to one of the many introductory texts, including Wasserman and Faust (1994), Prell (2011), Knoke and Yang (2008), Scott (2000), de Nooy, Mrvar and Batageli (2005), van Duijn and Vermunt (2006), and Hanneman and Riddle (2005), for a more detailed introduction. We assume that the reader is familiar with standard statistical techniques such as regression analysis and logistic regression. Kolaczyk (2009) and Knoke and Yang (2008) both have sections introducing ERGMs. The former provides a good technical and comprehensive treatment of statistical models for networks, which might be useful as further reading for the purposes of introducing alternative models. Reference is occasionally made to more advanced issues in statistics that the interested reader can follow-up on if a more comprehensive picture is desired.

Section I

Rationale

What Are Exponential Random Graph Models?

Garry Robins and Dean Lusher

2.1 Exponential Random Graph Models: A Short Definition

Exponential random graph models (ERGMs) are statistical models for network structure, permitting inferences about how network ties are patterned. Put another way, ERGMs are tie-based models for understanding how and why social network ties arise. This focus aligns ERGMs with a principal goal of much empirical social network research, which is to understand a given “observed” network structure (i.e., a network on which a researcher has collected data), and so to obtain insight into the underlying processes that create and sustain the network-based social system.

Much of social network analysis has been concerned with representing the network, a graph G , through various summary measures. From the literature, the reader may be familiar with summary measures $z(G)$ such as the number of edges in G , the number of mutual ties, centrality measures, triad census, and so on. We call these summary measures “network statistics,” and in mathematical terms, the ERGM assigns probability to graphs according to these statistics:

$$P_{\theta}(G) = ce^{\theta_1 z_1(G) + \theta_2 z_2(G) + \dots + \theta_p z_p(G)}.$$

The probability of a given network G is given by a sum of network statistics (the z s in this expression) weighted, just as in a regression, by parameters (the θ s) inside an exponential (and where c is a normalizing constant). The network statistics are counts of the number of network configurations in the given network G , or some function of those counts. These configurations are small, local subgraphs in the network. In short, the probability of the network depends on how many of those configurations are present, and the parameters inform us of the importance of each configuration.

This expression is explained in much more detail in Section II. However, because the mathematical features are not important for our purposes here, we hope to explain ERGMs in a relatively intuitive way in this introductory section.

To put it as simply as possible, a researcher specifies an ERGM by choosing a set of configurations of theoretical interest. As we will see, there are many sets of plausible configurations that can be used. Then, by applying this particular model to an observed social network, parameters are estimated. This permits inferences about the configurations – the network patterns – in the data, and this in turn allows inferences about the type of social processes that are important in creating and sustaining the network. Thus, ERGMs provide a methodology to investigate network structures and processes empirically.

Note that there is not just one ERGM – there are whole classes of them. The researcher has to choose the specification of an ERGM for the data (just as a researcher has to choose the variables to include in a regression). For an ERGM, the specifications involve choices of configurations that the researcher believes are relevant to the network structure. Although there are some standard ways to do this, the choices are ultimately based on theories about how ties come into being and appear in regular patterns. We discuss some of these theories in greater length in [Chapter 3](#). However, an ERGM itself carries some metatheory about networks, a conceptualization of a social network, and how it is created.

2.2 ERGM Theory

The use of an ERGM is consistent with some basic theoretical assumptions about social networks:

- Social networks are locally emergent.
- Network ties not only self-organize (i.e., there are dependencies between ties), but they are also influenced by actor attributes and other exogenous factors.
- The patterns within networks can be seen as evidence for ongoing structural processes.
- Multiple processes can operate simultaneously.
- Social networks are structured, yet stochastic.

Each idea is discussed more extensively in the following chapters. What is evident is that ERGMs are not “theory free.” The preceding list details certain claims consistent with the use of ERGMs that may shape the sort of social network questions that can be asked. Yet many of these claims are not exclusive to ERGMs and overlap with other thinking about social networks. For instance, an ERGM views social tie formation as locally

constructed, and much social network theory refers to local processes (e.g., reciprocity, transitivity, homophily – see [Chapter 3](#)). On the issue of multiple processes, Monge and Contractor (2003) argued that network research should be multitheoretical – that is, research should examine multiple theoretical perspectives at the same time, and, indeed, ERGMs can do this (see Section 3.1.5).

An explicit and particularly important feature of ERGMs is that network ties depend on one another (i.e., there is network self-organization); thus, the presence of one tie may affect the presence of others (see Sections 3.1.4 and 3.2.1 for general details, and Chapters 6 and 7 for more specific details). Of course, it is well understood that within a social network, individuals are by definition interdependent. In a seminal social networks article, White, Boorman, and Breiger (1976) famously lamented the divergence of theoretical and methodological perspectives on social interaction. On the one hand, they argued, theories are largely concerned with interaction; however, on the other hand, in empirical practice, researchers revert to aggregating individuals by categories. Thus, it is insufficient when we analyze such networks to consider individuals as unrelated “units of analysis” and instead more realistic to consider them as “actors in social relations” (Abbott, 1997, 1152). ERGMs take interdependency one step further by supposing that there is interdependence between network ties. It is in this way that network ties are construed as forming important patterns – the “configurations” parameterized in an ERGM. It is a theoretical and empirical task to delineate the various forms of dependence that are exhibited in actual social structures. We regard this as social network theory at a fundamental level.¹ Dependency among network ties is discussed at many stages of this book, particularly in [Chapter 7](#).

Although there are certain assumptions that one engages when using ERGMs, there is also considerable freedom available to the researcher. As noted previously, there is not just one ERGM, but many. A researcher chooses which model to use by selecting which network structures are important. The utility, as well as the challenge, of ERGMs is the specification of a particular theory (or theories) into social network terms – specifically, in articulating network configurations that, in isolation or combination, reflect a relevant theoretical concept. We hope it is clear

¹ We are referring here to the particular way in which dependence is modeled for cross-sectional data. There are a number of other cross-sectional statistical models that also cater to dependency-related issues such as the random effects p_2 model (van Duijn, Snijders, & Zijlstra, 2004) and latent social space/variable models (Airoldi et al., 2008; Handcock, Raftery, & Tantrum, 2007; Hoff, Raftery, & Handcock, 2002; Schweinberger & Snijders, 2007). The ERGM approach to dependency, however, is arguably the first and most explicit. See Snijders (2011) for an overview of statistical approaches to network analysis.

from Section III how this might be done. The process of theory translation requires the alignment of theoretical concepts with network configurations. As examples of how this is done, Chapters 14 and 17 focus on the actor attributes that are associated with receiving many network ties, whereas [Chapter 15](#) considers endogenous network processes – specifically, on different closure mechanisms. Yet the attribute-network tie associations in Chapters 14 and 17 are modeled while controlling for endogenous (i.e., self-organizing) network processes, including closure. Likewise, [Chapter 15](#) models endogenous network processes while controlling for network effects related to actor attributes. There are commonalities in the network structures examined in all three chapters (e.g., reciprocity and in-degree effects), demonstrating that many theoretical concepts relating to social tie formation are being analyzed simultaneously in different studies and highlighting that multiple theories need to be considered simultaneously. However, the analyses across the three chapters by no means engage with identical theories. Many theories may be examined in an ERGM, but some may be more important to the particular question in which a researcher is interested. In summary, an ERGM is a relatively open framework that is amenable to the testing of a range of network theories, depending on the interest of the researcher. Once a researcher has translated social theory into hypotheses in relational, quantitative terms, the ERGM provides a framework within which these hypotheses can be statistically examined.

2.3 Brief History of ERGMs

Moreno and Jennings (1938) introduced statistical network approaches by comparing observed network data to that expected under a null distribution. Although their method was simple enough, it did provide the insight that structural effects introduce “bias” into randomness. Some years later, this was picked up quite explicitly in the biased net theory of Rapoport (1953, 1957). Models for randomness itself were provided by the Erdős-Rényi graph (Erdős & Rényi, 1959) or, equivalently, the uniform Bernoulli graph distribution (Frank, 1981). (A number of refinements of null distributions have since been proposed in the literature – see Pattison et al., 2000.)

Holland and Leinhardt (1981) introduced a dyadic independence statistical model called the “ p_1 model,” which was the first ERGM to extend these simple random graph distributions. Although they set the grounds for future work, they did not go beyond dyads. Their simple model could be estimated using standard log-linear models; however, once triadic formations came into play, the independence assumptions necessary for log-linear models were expressly undermined. (More sophisticated

dyadic-based models, with extradyadic dependence, have since been proposed by van Duijn et al. (2004).)

The crucial work from which all else flows, including this book, was that of Frank and Strauss (1986). They decided that rather than searching for new clever methods based on independence, a proper approach to network dependence was required. Translating and extending spatial statistical approaches into a network context, they proposed Markov random graph models that became the mainstay of ERGMs for nearly two decades. This was indeed a landmark paper in the history of ERGMs.

Yet the Markov random graph approach was slow to be adopted by network researchers. It was not until the 1990s that an influential article by Wasserman and Pattison (1996) popularized this type of network modeling as p^* models (with the nomenclature p^* following p_1 – see also Rennolls, 1995). The emphasis put on the log-linear form of the p^* facilitated extensions of the basic Markov graph framework that resulted in models for multivariate (Pattison & Wasserman, 1999), valued (Robins, Pattison, & Wasserman, 1999), and bipartite (Skvoretz & Faust, 1999) network data. Actor attributes were incorporated in social selection (Robins, Elliott, & Pattison, 2001) and social influence (Robins, Pattison, & Elliott, 2001) models. ERGMs were proposed as an effective vehicle for investigating theories in network social science (Contractor, Wasserman, & Faust, 2006; Monge & Contractor, 2003).

These developments, however, were limited by the estimation procedure (maximum pseudolikelihood estimation) that did not properly deal with the assumed dependencies in data. In the past decade, nevertheless, the pace of new development for, and the popularity of, ERGMs has grown dramatically. Various groups worked on algorithms to calculate (or, more precisely, approximate) maximum likelihood estimates, all being based on computer-intensive simulation (Corander, Dahmström, & Dahmström, 1998, 2002; Crouch, Wasserman, & Trachtenberg, 1998; Handcock, 2002, 2003; Snijders, 2002). What at first seemed to be algorithmic difficulties were suspected by Handcock (2002) and Snijders (2002) to be deficiencies of the Markov specification of the ERGM, which up to then was considered the natural default specification capturing tendencies toward transitivity, as proposed by Frank and Strauss (1986) (for model deficiencies, see Section 6.6.3 on the Markov model and Section 12.6 on degeneracy and near-degeneracy). As a remedy to these deficiencies, and building on the realization-dependent conditional independence assumptions developed by Pattison and Robins (2002), the Markov specification was generalized to specifications satisfying the social circuit model in Snijders et al. (2006). This proposal was a breakthrough in improving the capacity of ERGMs to fit real data (Robins et al., 2007).

A number of simulation-based algorithms for approximating the maximum likelihood estimate were implemented in software. This led to three

publicly available computer programs: SIENA (Snijders et al., 2005), in which the ERGM estimation was included as an extension and algorithmic analogue of the previously available estimation methods for longitudinal actor-oriented models; the *statnet* software suite (Handcock et al., 2008), which used a different algorithm to produce the same estimates; and the PNet package (Wang, Robins, & Pattison, 2005), which used the same algorithm as SIENA.

There were ensuing steps in model formulation and estimation (Hunter & Handcock, 2006) and in novel goodness-of-fit approaches (Hunter, 2007; Hunter, Goodreau, & Handcock, 2008; Robins et al., 2007). As described later in this book, new developments have since begun to proliferate. These include approaches to estimation under varying circumstances – such as Bayesian inference and models for missing data and snowball sampling – as well as new models for directed, bipartite, multivariate, longitudinal, selection, and influence data. These extensions are described in Section II of this book.

2.4 Network Data Amenable to ERGMs

An ERGM is designed to be used with empirical network data. As we will see, it is possible to simulate networks using ERGMs (see Chapters 12 and 13), but the strength of the method comes to the fore with observed network data, the data that researchers have collected and are keen to understand.

Section III of this book highlights a number of different types of networks that are amenable to analysis using ERGMs. Networks can be directed or undirected,² and all networks used with ERGMs are binary. Valued network ties are typically not used except as covariate, exogenous networks to predict the binary network. ERGMs have been formulated for ordinal ties (e.g., strong/weak – Robins et al., 1999), but this has yet to be implemented in current estimation software.

An ERGM is typically used with completely observed networks; however, this is not to say that it could not be used with ego or personal networks. As noted in Chapter 12, network data derived from snowball sampling can also be estimated with ERGMs, and estimation can proceed even when some network data are missing.

The number of nodes is an important consideration because estimation processes can be computationally intensive (and assumptions of homogeneity may be tenuous in large networks). With current computing capability, estimation of networks of more than 1,000 to 2,000 nodes

² Throughout the book, we use the terms “undirected” and “nondirected” interchangeably.

is possible (Goodreau, 2007). The new snowball sampling techniques, however, open future possibilities of estimating very large networks.

Most ERGMs to date have been applied to cross-sectional data, but longitudinal ERGMs are available as described in [Chapter 11](#). Typically, ERGMs examine one type of network tie, although multiple networks can be estimated simultaneously ([Chapter 10](#)), and models can include other networks as exogenous covariates ([Chapter 8](#)). ERGMs are also available for bipartite network data ([Chapter 10](#)) and, in principle, can be developed for any sort of relational data.

Multilevel analyses can be performed with ERGMs (Lubbers & Snijders, 2007), whereby ERGMs can be estimated for a number of similar networks (e.g., networks in several school classrooms), and then multilevel post hoc analyses conducted to assess significant parameters across the entire data set.

It is possible to focus solely on the examination of network ties (and thus to disregard actor attributes) with ERGM models. However, ERGMs can also incorporate any number of binary, categorical, and continuous actor attributes to determine whether actor attributes are associated with the formation of network ties ([Chapter 8](#)), and geospatial variables can also be included in an ERGM ([Chapter 8](#)). Finally, it is possible to use the general ERGM approach for models that predict the presence of binary actor attributes given the presence of network ties. These autologistic actor attribute models (ALAAM) can be used to examine a variety of social influence and related effects ([Chapter 9](#)).

Formation of Social Network Structure

Dean Lusher and Garry Robins

3.1 Tie Formation: Emergence of Structure

It is not always recognized that an exponential random graph model (ERGM) carries theory in the form of assumptions about networks, network processes, and social structures. We can think of “ERGM theory” as network metatheory because it is not specific to a particular network process. It is a theoretical perspective within which more specific network theories may be investigated. The essence of ERGM theory is the formation of social structure through the accumulation of small local substructures and, ultimately, through the formation of individual ties into the patterns of those substructures.

3.1.1 Formation of Social Ties

ERGMs are first and foremost concerned with explaining the patterns of ties in a social network. This tie-based approach of ERGMs permits answers to some questions but not others. A standard ERGM is *not* a model focused on predicting outcomes for individuals in the network (so called diffusion or social influence models¹); instead, it is about revealing patterns that may enable inferences on tie formation processes, including social selection processes where network ties are predicted from the attributes of the network actors.

Of course, there are many network theories about tie formation or tie patterning that can be drawn on in specifying an ERGM for a given network. For instance, reciprocity or exchange is seen as a basic and universal human activity (Blau, 1964), so that, generally, in human social networks we expect ties to be reciprocated. Beyond dyads, the importance

¹ An ERGM-type approach can be used for social influence models, however (see [Chapter 9](#)).

of triadic relations was proposed by Simmel (1950). Following Heider (1958), Cartwright and Harary (1956) introduced structural balance theory, providing a graph theoretical approach to triangulation among social network ties, otherwise known as “path closure” or “network closure.” Since then, network closure has become a central theme in social network research. Granovetter (1973) contrasted closure of strong ties with nonclosure of weak ties. Burt (1992) examined network brokerage and structural holes, suggesting that the person in the center of a nonclosed structure is advantaged. In contrast to closure, theories of prominence in networks also suggest that people who are socially well connected may be advantaged or have distinctive status (Bavelas, 1950; Freeman, 1977, 1979). Preferential attachment describes how network popularity may induce further popularity (Barabási & Albert, 1999; Merton, 1968; Yule, 1925), resulting in some high-degree actors in the network. In regard to actor attributes, the importance of homophily has been well documented (McPherson, Smith-Lovin, & Cook, 2001) as a means to explain the presence of network ties. For multiple networks with different types of tie, Nadel (1952) and White (2008) suggested one social network could provide a context for another, so that there may be interdependencies *across* different networks, whereby ties in one network might encourage the formation of ties in another.

3.1.2 Network Configurations: Consequential Network Patterns and Related Processes

These theories and ideas provide explanations as to why ties might be present in a network, how ties might come to form particular local network patterns, and how ties might be associated with actor attributes. In terms of an ERGM, we call such local network patterns “network configurations.” A network configuration is a possible small subgraph that may represent a local regularity in social network structure. It is an empirical question whether a particular configuration is present in a given network, but configurations embody some ideas about how networks may be patterned locally.

Some illustrative examples of network configurations for directed graphs are given in Figure 3.1. In the first example on the left, the double-headed arrow between two nodes represents the reciprocation of ties between two actors. The next configuration represents transitive closure in the form of a particular triadic structure (note that there are other directed triadic forms depending on the direction of the arrows and whether some of the ties in the triad might also be reciprocated). The next configuration (labeled “Activity”) is termed an “out-2-star”: it is a starlike structure with two outgoing ties from the central node. Outgoing starlike structures are often used to represent activity-based configurations, where

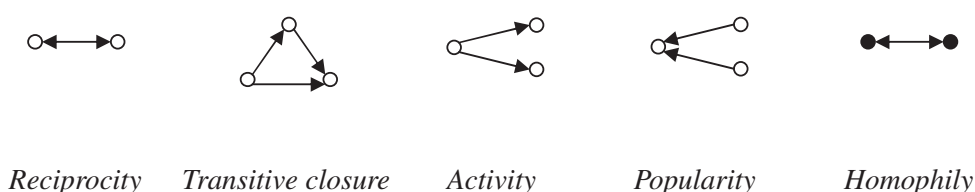


Figure 3.1. Some network configurations and their underlying social processes.

an actor directs ties to many network partners. These configurations obviously relate to the out-degree distribution of the graph. There are also in-star configurations, often described in terms of network “popularity,” that relate to the in-degree distribution. Finally, the fifth configuration is a structure where two actors with the same attribute (e.g., gender – hence the similar coloring on the nodes) have a reciprocated tie. This is often used to represent reciprocated homophily, whereby ties are associated with similar actor attributes.

Social networks are often seen as emerging from various social processes or mechanisms (Hedström & Swedberg, 1998), in which case the patterns of network ties can be revealing about the processes that give rise to them. The idea of process invokes the dynamics of the system, perhaps suggesting that we need longitudinal observations (as discussed in [Chapter 11](#)). Yet, structural processes will result in network patterns that are evident even in cross-sectional data. The presence of these configurations (such as in [Figure 3.1](#)) might be thought of as akin to archeological traces that are etched into network structure by social mechanisms operating across time. For instance, suppose that people tend to collaborate more if they share (say) the same profession, the well-known network process of “homophily”. If we observed the system at multiple time points, we would see a series of collaborations forming between pairs of individuals who share professions. At the last (cross-sectional) observation point, we will see more of these same-profession collaboration links than we expect to see by chance, regardless of whether we take heed of the earlier observational points. The final cross-sectional observation in itself provides evidence for homophily because we will see more of the homophily configuration of [Figure 3.1](#) than we would expect just by chance. So, even using cross-sectional network data, by examining appropriate network patterns we can obtain insight about the social mechanisms that drive the structure of the network.

The importance of a network configuration is that it is a consequential pattern that may represent an underlying social process.² This

² Of course, there are some networks for which this statement is not applicable, especially those that are not social networks. ERGMs can still be applied to these networks, but the interpretation of network configurations will then differ.

explanation maintains a distinction argued by Harrison White (2008) between a network structure and the process by which it has arisen, implying that “a network structure is the consequence of a dynamic process”.³ Of course, this distinction between pattern and process is not limited to ERGMs, but relating static structures to social processes is particularly important to ERGMs even though it is often cross-sectional data that are modeled. (Mathematically, it can indeed be shown that the ERGM is the end product of a process of tie formation and deletion – see Section 11.3.4.)

3.1.3 Local Network Processes

In ERGMs, the configurations we study are local. A network tie is between a pair of individuals. If we ask how network ties come into being, we need to focus on pairs, not on individuals one at a time. So, an ERGM is a “tie-based” model for social networks.⁴ A tie comes into place in response to the existing local social environment within which the two individuals operate. This social neighborhood contains other pairs of actors and their ties. Thus, we are not concentrating only on pairs of individuals, even if that is our starting point. We may have quite complex local network structures involving several ties in the local social environment. The patterning of these other ties, in addition to the properties of actors in the local neighborhood, may influence the presence of a new tie. In this way, network configurations can come into being.

3.1.4 Dependency (and Theories of Network Dependence)

A fundamental concept underpinning an ERGM is of the dependency between network ties. Without some form of dependence among ties, it is impossible to argue for tendencies for certain patterns of ties to form. So, to postulate configurations as consequential network patterns is to postulate dependence among network ties. If ties do not depend on each other – in the sense that the presence of some ties influences the presence of others – then there is no impetus for them to form configurations.

Importantly, to make the local configuration approach work, we need a theory of what counts as “local” and what specific configurations to look for. Perhaps surprisingly, it turns out that the answer to these two questions is one and the same. To formulate an ERGM, we need a theory

³ It is also true that process can be constrained by structure.

⁴ Although an ERGM is not an “actor-based” model where decisions of individual actors are modeled explicitly in terms of preferences and constraints, as in the stochastic actor – oriented models for longitudinal networks of Snijders (2002), an ERGM is not inconsistent with an actor-based theory. In Chapter 7, we see also how ties depend on each other through individual actors. For explicit links between actor-oriented and edge-based processes, see Section 11.4.2 and Snijders (2006, 2010).

of “dependence,” and fortunately, there are now several available for a researcher to consider. Once we adopt a theory of dependence, it turns out that we automatically have a definition of “local,” and the types of patterns to consider are then constrained. However, if we believe that certain network patterns are important based on more specific social science/social network theory, we are adopting (implicitly or explicitly) a particular dependence hypothesis and a consequent definition of “local.”

There are a number of possible dependence assumptions that are set out more fully in Chapters 6 and 7. Each has a different definition of what counts as a local social neighborhood, and so implies a different family of ERGMs. Recent work on how to understand these different definitions, as well as the relationships among them, is described in [Chapter 22](#) of this book.

As a simple example, suppose someone in your workplace often confides in you about important issues. This is likely to increase your tendency to communicate with that person. We know this as reciprocity, an important feature of social life. Reciprocity is a form of dependency, whereby the two possible directed ties within a dyad are dependent on each other. In other words, the presence of tie from Fred to Mary increases the chances of a tie from Mary to Fred (and vice versa). This is known as “dyadic dependence.” The reciprocal tie configuration was presented in [Figure 3.1](#). Here, a dependence hypothesis (dyadic dependence) naturally leads to a decision about a configuration to apply in an ERGM (reciprocity).

Previously, we described dependence in terms of local social neighborhoods that affect the presence of network ties. If dyadic dependence is the only applicable dependence assumption, then the definition of “local” is simply the dyad. There are no other effects outside the dyad that affect the presence of ties (hence, it is sometimes referred to as “dyadic independence” because the dyads are independent of each other).

Of course, there can be more complex dependencies instead of, or in addition to, dyadic dependence. As a second example, the chances of any one individual being a friend of President Barack Obama are small. However, if you are friends with Michelle Obama, then your chances of striking up a friendship with Barack substantially increase. We know of this as transitivity or path closure, or “a friend of a friend is a friend.” In later chapters, we see how this configuration (along with others) arises from “Markov dependence,” where ties are assumed dependent if they share a node.

Still other dependency assumptions are possible. Consider, for example, the presence of two couples at a party. Because one partner from each couple may be friends and start talking away, their previously unacquainted partners may strike up a conversation with one another and over time become friends. The presence of existing relationships creates

the conditions whereby an old friendship tie affects the chances of a new friendship. We know this pattern of relations as a “4-cycle,” and it can be an important configuration arising from “social circuit dependence.”

In addition, there are particular types of dependence assumptions relating to actor attributes and dyadic and extradyadic covariates of network ties.

In these cases, we see how dependency among social ties can give rise to network configurations. Reciprocity, transitive closure, and four-cycles can come about because the formation of a tie depends on the presence of other ties in certain ways. Dependence helps us define what is local, but it also gives us an implied explanation of dynamics.

However, standard statistical approaches assume *independence of observations*. From an independence perspective, if someone communicates with you, this will have no bearing on whether you communicate back. From an independence of observations perspective, you assume that being friends with Michelle Obama does not affect your chances of being friends with Barack, and previously unacquainted husbands have no greater chance of coming to know each other, even if their wives are already friends. These do not seem reasonable descriptions of social life. So, if we are to understand systems based on some form of dependence between ties, we cannot rely on standard statistical methods such as *t*-tests, analysis of variances (ANOVAs), and regressions that assume independence. Because an ERGM can incorporate dependence, it is a more principled approach to understand the patterning of network ties.

3.1.5 Complex Combination of Multiple and Nested Social Processes

It is important to note that the tie-based approach of ERGMs permits an understanding of the “complex combination” of social processes by which network ties are formed. We point to multiple and nested processes.

Multiple Origins of Social Network Structure. Previously in this chapter, we presented a number of theories about the formation of network ties, without attempting a complete list. However, it would be a brave person who would suggest that one and only one of these processes explains *all* there is to know about the organization of a social network. The totality of ties in a network is not likely to be explained only by homophily, or only by reciprocity, but it is certainly feasible that both processes may be at play at the same time within the one network. By incorporating a number of configurations simultaneously into a model (such as one for homophily and one for reciprocity), an ERGM can test the evidence as to which processes contribute to the formation of the network structure (Monge & Contractor, 2003).

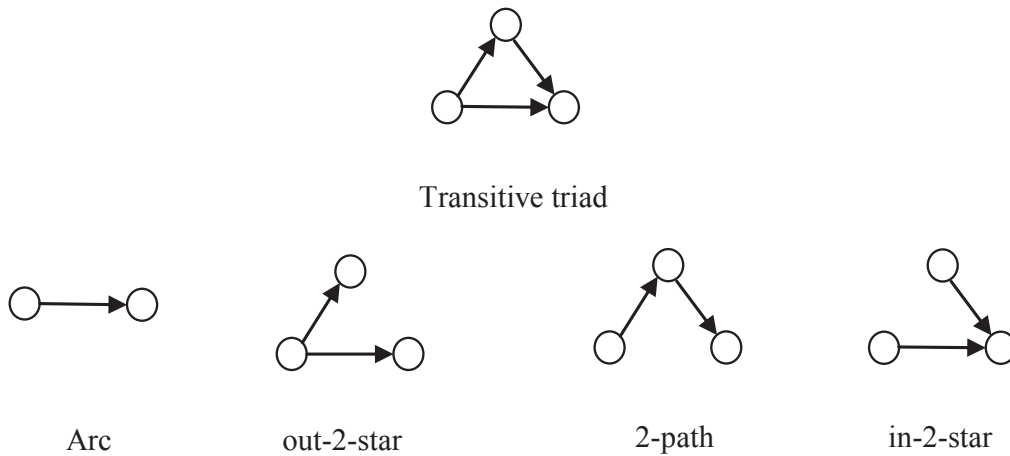


Figure 3.2. Nested configurations for a transitive triad.

In the one social network, there is no a priori reason why multiple social processes should not be present. Because humans are intentional beings with multiple motivations for and multiple expressions of social action, it is especially in human social networks that we expect that multiple processes will occur simultaneously. Of course, we do need to be guided by theory, and we certainly need to be empirical, but expecting a simple explanation for a complex human social system is naive.

Nested Configurations. It is important to realize that configurations are often nested within one another. The simplest network configuration is a single network tie. Every other configuration obviously contains this configuration, and possibly others. Consider a transitive triad. This configuration includes within it three single arcs: one 2-path, one out-2-star, and one in-2-star (see Figure 3.2 for a depiction of these configurations).

Because configurations are nested in one another, it is not enough, for instance, to observe many triangles to infer an effect for network closure. There may be many triangles because there are many ties (i.e., the network is dense), or because there are many 2-paths or 2-stars (i.e., in- and out-2-stars for directed networks). To have solid evidence of network closure, we need to observe more than expected triangles given (i.e., taking into consideration) the number of 2-paths (and arcs and 2-stars) in the data.

Given an average baseline propensity to form arcs, if that propensity is strong enough, we will see some transitive triads simply by chance. The stronger the baseline effect, the more transitive triads. The same applies if there are propensities to form 2-stars and 2-paths. Thus, we can only infer a specific process of transitive closure by considering the propensities for the four lower-order configurations and a transitivity effect *simultaneously*. We ask, do we see more transitive triads in the

network data than we expect to see *given* the average tendency to form arcs, 2-stars, and 2-paths? If so, we have evidence for a transitive closure effect. If not, then the presence of transitive triads in the network can be explained more simply by the presence of the lower-order configurations, and we do not need to postulate a closure effect to explain this network structure.

In summary, when we talk of multiple processes, it is not just that there are many choices. Rather, local, multiple, and nested network effects combine into a complex combination of processes. ERGMs are about understanding the complexity of a social system – its multiplicity, its interconnectedness, and its dependencies.

3.2 Framework for Explanations of Tie Formation

In this book, we discuss three broad categories of tie formation processes, represented in [Figure 3.3](#): self-organizing network processes, attribute-based processes, and exogenous dyadic covariates. More specific effects can be identified under each of these three broad categories. [Figure 3.3](#) does not claim to be exhaustive or exclusive because, of course, there are many possible tie formation processes. For instance, exogenous contextual factors could include categories for culture or settings and so on. Rather, [Figure 3.3](#) is a schematic that focuses on topics of interest in the ensuing chapters. Each process here, however, has a well-established basis in social network or social science theory. Each results in networks with particular types of configurations.

3.2.1 Network Self-Organization

Network ties can organize themselves into patterns because the presence of some ties encourages others to come into existence. We often refer to these as “purely structural” effects because they do not involve actor attributes or other exogenous factors. They are “endogenous” effects in that the network patterns arise solely from the internal processes of the system of network ties.

Network self-organization can arise through degree-based effects. For instance, individuals – by being popular – may attract even more popularity. In that case, we expect diversity in popularity across the actors, which translates into higher dispersion in the degree distribution, or equivalently, into greater network centralization with network ties directed to a few highly central nodes. In social network theory, such processes are often referred to as “preferential attachment” (Barabási & Albert, 1999) or the “Matthew effect” (de Solla Price, 1976). There are several different versions of degree-based processes: for instance, in directed networks,

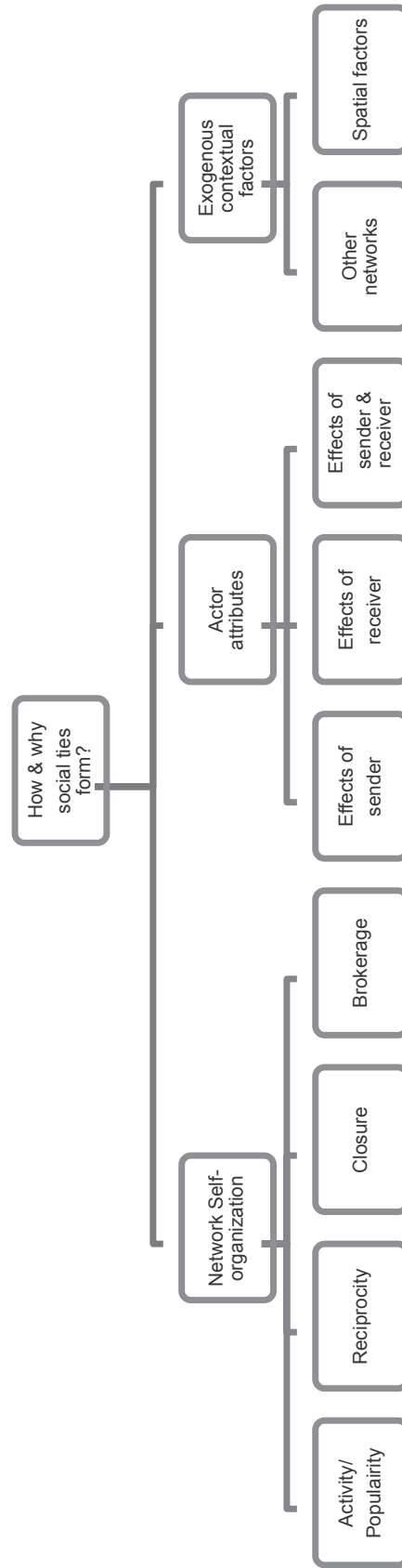


Figure 3.3. Conceptual framework for processes of social tie formation discussed in this book.

it is possible to have effects related to both in-degree (“popularity”) and out-degree (“activity”). An example configuration relating to in- and out-degrees are the in- and out-2-stars depicted in Figure 3.2. A node with high out-degree is centered on many out-2-stars, so networks that are highly centralized in terms of out-degrees (activity) will have a high number of out-2-stars; similarly for in-2-stars and in-degrees (popularity) (the number of 2-stars is directly related to degree variance and centralization (Hagberg, 2004; Snijders, 1981a, 1981b)). A “2-path” or “mixed-2-star” is also possible where a node has one network tie directed toward it and one directed away from it (Figure 3.2), reflecting correlations between in- and out-degrees (e.g., good listeners who do not talk too much). These variations are possible in directed networks; in undirected networks, we have only the 2-star configuration because the difference between the three types of directed 2-stars disappears. Of course, it is also possible to have higher-order star configurations – 3-stars, 4-stars, and so on – with three or more ties centered on the one node (these are, in fact, moments of the degree distribution and reflect skewness and kurtosis, respectively).

A third important self-organizing effect involves network closure. Especially in networks invoking some form of positive affect or collaboration, as we discussed previously, there are often tendencies for 2-paths of actors to “close the path” by forming a third tie that produces a triangle (Davis, 1970). This is also referred to as “network clustering,” or in directed networks, “transitivity” (when the tendency is for the formation of transitive triads).

Triangulation reflects the human social propensity to operate in group-like structures. A network triangle involving three ties can be viewed as a simple archetypal expression of a small group, in the sense that a group is likely to appear as a subset of actors with many ties among them. In that case, we will see many triangles among that subset. Thus, over and above the formation of individual triangles, there is often a process of “multiple triangulation,” where triangles occur together among larger subsets of actors in clique-like structures rather than as separated features spread evenly across the network (akin to *community structure* – see Newman and Park (2003) – or other long-standing definitions of cohesive subgroups – see Wasserman and Faust (1994)).

There are several different versions of triangulation and network closure. A “transitive triad” configuration for directed networks is in Figure 3.2. A transitive triad represents network closure of a particular type because it implies a local hierarchy, with one node receiving two ties and sending none, one node receiving no ties and sending two, and one node receiving and sending a tie. Depending on the content of the network relationship, in the right circumstances this may be interpreted as the receiving node being the most popular within the group-like triangular structure. If instead we have a “cyclic triad,” where the direction

of all ties is consistent so that they form a “3-cycle,” then none of the three nodes would be singled out. Of course, in undirected networks, there is no distinction between a transitive triad and a cyclic triad: only an undirected triangle is possible.

Under network self-organization, [Figure 3.3](#) also includes provision for “network brokerage,” which has been a major theme in social network research in recent years (e.g., Burt, 1992). This has been the topic of recent ongoing work in the specification of ERGMs and is presented briefly in [Chapter 6](#) (see Section 6.7).

3.2.2 Individual Attributes

Individuals bring their own capacities, capabilities, and predispositions to a social system. The qualities of individuals in networks can be very important to the formation of network ties (Emirbayer & Goodwin, 1994; Kilduff & Krackhardt, 2008; Parkhe, Wasserman, & Ralston, 2006). Depending on the domain of research, many types of individual-level variables – basic demographics such as age or gender, expertise or profession, and psychological motivations and attitudes – may be relevant. In network terms, these individual measures are termed “actor attributes.” For ERGMs, we often use the terminology of “actor-relation effects,” which refers to “the association of a particular attribute with a social tie,” representing the propensity of actor attributes to affect tie formation.

An actor attribute may operate by affecting the involvement of the actor in the network. For instance, in directed networks, an attribute might encourage that individual to be more *active* (i.e., express more ties) or make that actor more *popular* (i.e., receive more ties). We call these the “sender effect” and the “receiver effect,” respectively. Importantly, these effects are different to preferential attachment mentioned previously. Preferential attachment suggests that new ties occur because of the presence of other ties. A sender or receiver effect suggests, however, that actors send or receive more ties because of their attributes. So, in an ERGM, we can delineate popularity (and/or activity) as resulting from network self-organization or actor attributes.

Because ties involve pairs of actors, attributes of both parties can also come into play in the formation of network ties. For instance, an individual might select a partner because their attributes match. This process of homophily arises when people tend to be connected to others who are in some way similar to themselves.

Thus, there are several distinct processes whereby attributes may affect network ties, operating at the level of a single actor or across two actors. There is a need to consider processes together to make the right inference from the data. Suppose that, in a school, girls tend to be more active

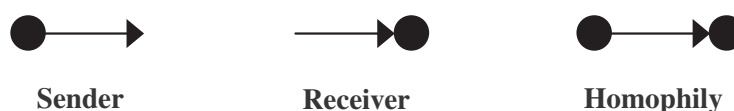


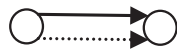
Figure 3.4. Examples of network configurations for actor-relation effects.

than boys in having friends. Then, simply by chance, there will be a higher proportion of friendship ties between pairs of girls because girls are implicated in more ties anyway. If all we do is to observe the many ties between girls, we cannot conclusively infer that there is a tendency for homophily over and above a simpler activity or popularity effect. It is only by considering effects simultaneously that we can disentangle whether there is a predisposition for girls specifically to select other girls as friends.

In an ERGM, actor attributes are treated as exogenous or explanatory variables that affect the presence of social ties. These are discussed as social selection models in [Chapter 8](#). (It is also possible to have models that predict attribute status, with network ties as exogenous predictors – see [Chapter 9](#).) Network self-organization, in contrast, is treated as an endogenous process. The distinction between endogenous and exogenous explanations for the presence of social ties is important. We need to account for purely structural tendencies for tie formation in order to make the right inferences about actor attribute effects. If, for instance, we predict the presence of a network tie from actor attributes and assume that ties arise only through those individual qualities, we are not taking into account possible dependencies among network ties, and so we are ignoring network self-organization. As such, we may overestimate (or underestimate) the importance of attributes in the network.

Examples of network configurations for actor-relation effects are presented in [Figure 3.4](#). Here, a colored node represents an actor with a particular attribute status (e.g., “female”). Thus, in this example, the sender configuration represents a female expressing ties to other actors (irrespective of whether these other actors are male or female), the receiver configuration represents a female actor receiving ties from other actors, and the homophily configuration represents a tie between two females. Although this example presents configurations for a binary attribute variable, as explained in [Chapter 8](#), there are counterparts for continuous or categorical attributes.

Whether one specifically examines networks for actor-relation effects (as in [Chapter 14](#)) or uses them as controls when looking for network self-organization (as in [Chapter 15](#)), such effects are nonetheless important, and their addition to ERGMs is a significant advantage for testing network theory.



Entrainment

Figure 3.5. Example of network configurations for dyadic covariates.

3.2.3 Exogenous Contextual Factors: Dyadic Covariates

Finally, other exogenous contextual factors may be important to tie formation. In this book, we often treat these as dyadic tie covariates, although they do not necessarily have to be in only dyadic form. An instance of a dyadic covariate may be another social network considered to be fixed and exogenous to the model. An ERGM can then investigate whether the presence of the covariate tie predicts the corresponding tie in our network of interest (see [Chapter 8](#)). A good example is the importance of the formal organizational hierarchy on the presence of actual communication ties within an organization. Ties from the two types of network may co-occur (or be *entrained*), in which case, for instance, a worker may communicate with his or her boss. In that sense, an ERGM may help understand the extent to which top-down, centrally decreed structures relate to bottom-up informal networks.

A network configuration representing the dyadic covariate effect of entrainment is presented in [Figure 3.5](#). Here, the dotted arc represents the covariate network. The presence of these configurations in the network indicates the extent to which the network ties of interest co-occur with the covariate network ties. More elaborate dependencies on covariate networks may also be formulated, such as the multiplex configurations of [Chapter 10](#). As explained further in [Chapter 8](#), other covariates may also be relevant, including geospatial factors, shared affiliations, and contexts, and they can enter the models in distinctive ways.

In summary, the framework of [Figure 3.3](#) takes into account some of the many network ideas about tie formation and presents them to permit their understanding in terms of network configurations. This is the general structure whereby we formulate specific ERGMs to answer precise research questions about the origins of social network structure.

Simplified Account of an Exponential Random Graph Model as a Statistical Model

Garry Robins and Dean Lusher

This chapter provides a simplified overview of some methodological aspects of exponential random graph models (ERGMs), with the technical detail presented in Section II, specifically in Chapters 6 and 7. To begin, it is worthwhile to consider the value of a statistical model in understanding social network structure.

Harrison White made the important observation that “sociology has to account for chaos and normality together” (2008, 1). Social life is stochastic, and social networks are not predetermined or invariant. We do not expect that in a human social network, reciprocity will apply (strictly) in all situations; rather, there may be a tendency toward reciprocity in the sense that more reciprocation will be present than otherwise expected over and above what would result from other processes. In a sense, if we do not allow for “tendencies” with some variation, in the extreme, a nonstochastic model requires one unique explanation for each tie, present or absent.

Accordingly, it makes sense to use a statistical model such as an ERGM to investigate network structure. By incorporating randomness, statistical models deal with expected values, so we are then able to draw inferences about whether observed data are consistent with expectations.

The balance between randomness and order is an important issue in much social network research. For instance, in considering the “small world” nature of many social networks, Watts (1999) showed that adding a small amount of randomness to a highly structured graph could dramatically shorten path lengths. In an ERGM context, a configuration represents the ordered nature of local structure. If effects for configurations in a model are minimal (e.g., a tendency for reciprocity might be weak), then the resulting networks will be close to purely random. In contrast, if an effect is strong (e.g., a strong tendency for reciprocity), then the resulting networks will appear as highly structured (e.g., most ties will be reciprocated). Because an ERGM is stochastic, the model does

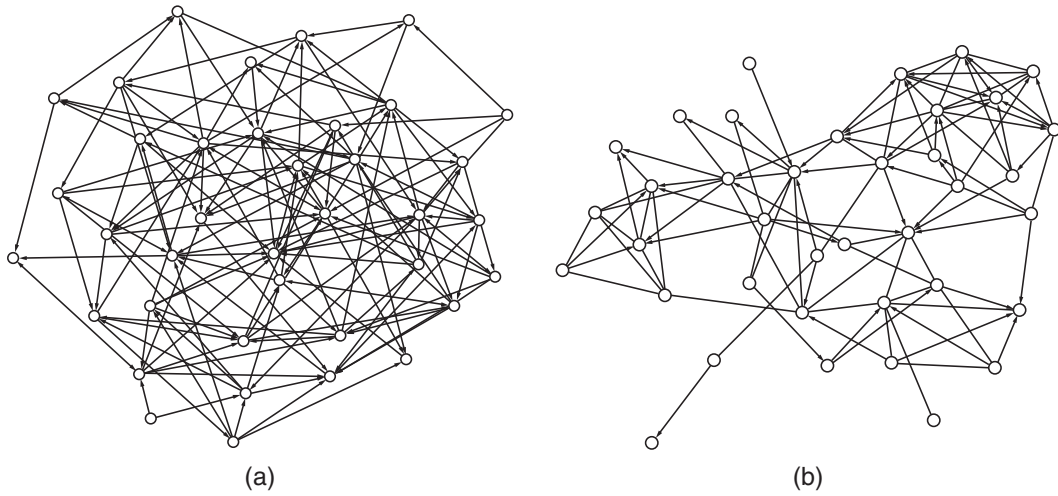


Figure 4.1. (a) Simple random network and (b) empirical communication network.

not imply just one network. The result is a “probability distribution of graphs,” which we discuss in more detail later in this chapter.

4.1 Random Graphs

We noted in [Chapter 3](#) that a given dependence assumption picks out certain configurations as important to the structure of the network. We have also noted that some configurations will occur just by chance, even in a network where ties are formed independently of one another (i.e., no dependence). [Figure 4.1a](#) presents a random graph of 38 nodes and 146 arcs generated by scattering the 146 arcs purely randomly among the nodes (to be precise, the graph comes from a uniform random graph distribution conditional on the presence of 146 arcs – see [Section 4.2](#)). The number of nodes and arcs has been chosen to match those of an empirical communication network in [Figure 4.1b](#). (This communication network is presented in more detail later in this chapter.) From the visualization, it seems that the random network is quite different from the empirical network. This also seems apparent from the selected network statistics for both networks in [Table 4.1](#).

From [Table 4.1](#), however, some network configurations still occur, even in the random network. When the 146 ties are distributed among the nodes by chance, there are still *some* reciprocated ties and *some* transitive triads but not nearly as many as in the communication network (although the in- and out-stars are not so different between the two networks). Nevertheless, although there are undoubtedly less reciprocated ties and transitive triads in the random network, how do we judge whether

Table 4.1. *Selected network statistics for networks in Figure 4.1*

	Random network	Communication network
Actors	38	38
Arcs	146	146
Reciprocated arcs	6	44
Transitive triads	53	212
In-2-stars	292	313
Out-2-stars	254	283

there are *significantly more* in the communication network than we might expect?

4.2 Distributions of Graphs

A statistical (nonparametric) technique to address whether there are significantly more particular network structures than expected by chance is to simulate a large sample of random (directed) graphs of 146 arcs on 38 nodes. This can be considered a sample of graphs from a “graph distribution” where every (di)graph is equally probable if it has 146 arcs (and has 0 probability otherwise). A graph distribution is simply the set of all possible graphs (in this case, on 38 nodes), with a probability assigned to each graph. The graph distribution we just described is called the U|L distribution (the uniform distribution of digraphs on 38 nodes given the number of arcs $L = 146$). (For further discussion of U|L, see Wasserman and Faust (1994).) It is a uniform distribution because each graph of 146 arcs has equal probability. Such approaches reflect the early history of statistical network methods, which was dominated by techniques relying on testing a hypothesis using a simplistic null distribution, going back to Moreno and Jennings (1938).

Having produced a large sample of graphs, we can then calculate the number of reciprocated arcs and transitive triads for each sampled graph. We did this for 1,000 simple random graphs with the results for reciprocated arcs presented in Figure 4.2. The mean of this sampled distribution is 7.5, and the number of reciprocated arcs ranges from around 0 to 15 across the 1,000 graphs. Accordingly, based on this information, if we had a network from this random graph distribution, we would expect to see around 7 to 8 reciprocated ties just by chance, and it would be very unusual to see more than 15 reciprocated ties. The communication network, however, has 44 reciprocated ties (represented in Figure 4.2 by the dotted vertical line), many more than expected by chance from a distribution of simple random graphs. This is evidence for a number of

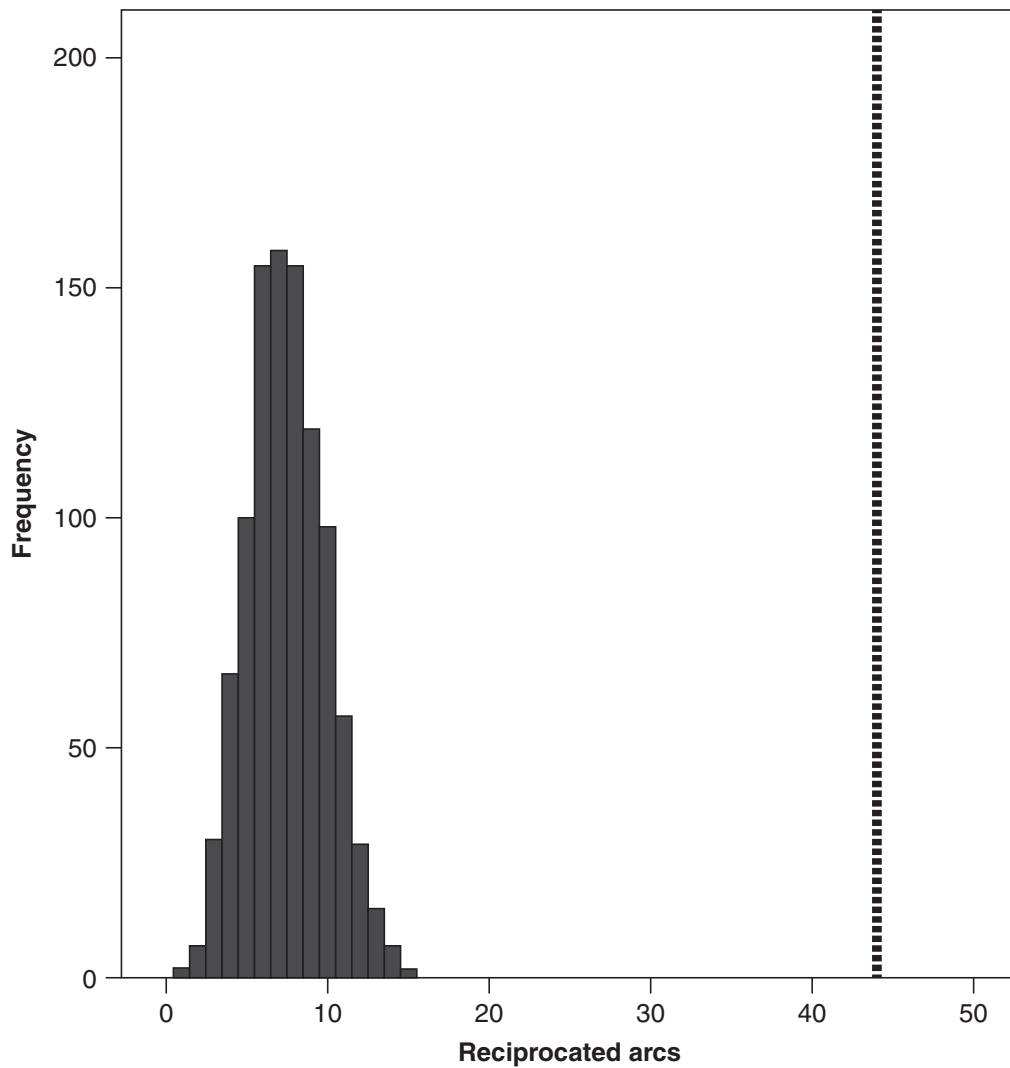


Figure 4.2. Distribution of reciprocated arcs from sample of 1,000 random graphs.

conclusions: (1) the observed network is not just a random graph (2) there is dependence among network ties, and (3) there are reciprocity processes active in this network.

To put the results more succinctly, the probability of observing 44 reciprocated ties in a random network is very close to 0. This is less than 5% (if we choose that as an alpha level); thus, in accord with standard null hypothesis significance testing, we can say that the number of observed reciprocated arcs is significantly above what is expected from the null distribution ($U|L$), and so there is significant reciprocation in this network.

We can draw a similar conclusion about transitive triads. From the sample of 1,000 simple random graphs, the mean number of transitive

triads is 55, with a standard deviation of 7.8. The observed number in the data of 212 is significantly greater.

However, there are two (related) limitations to this type of inference. First, our comparison is only to a distribution of random graphs (i.e., those with 146 arcs). Perhaps if we had another type of graph distribution (e.g., graphs with 146 arcs and 44 reciprocated ties), the number of transitive triads in the observed data would not be extreme. Second, in a related fashion, we only consider each effect one at a time, and we ignore the nesting of configurations. For instance, we can say that there are significantly more transitive triads, but we do not know whether that arises because there are an increased number of stars or because of a triangulation process. When we say that there are significantly more reciprocated ties and transitive ties than in the U|L distribution, we are rejecting the null hypothesis (the observed graph was generated by a U|L model), but we are not saying anything about the alternative – whether reciprocation or transitivity are significant processes.

What an ERGM tries to do, however, is to model the effects of interest (e.g., reciprocated ties, transitive triads) and, in relation to an observed network, to find a distribution of graphs where the observed data are central in that distribution. (More technically, we *fit* the model by *estimating parameters*.) In fact, for the communication network, an ERGM tries to find a distribution of graphs on 38 nodes, which have *an average* of 146 arcs, 44 reciprocated ties, and 212 transitive triads (and, indeed, also 313 in-2-stars and 283 out-2-stars to cater for nested effects). This might seem like a complex task and indeed it is. Fortunately for researchers, there is software for this purpose, so we do not necessarily need to know precisely how the algorithms work (although [Chapter 12](#) enlightens those who want to know). Later in this chapter, we demonstrate how to find this distribution of graphs for this particular network. In doing so, we discover which configurations are important in the network, which effects have independent explanatory value, and which can be explained by other effects.

In one sense, what an ERGM does is compare an observed network to the other possible ways it could be arranged. In all networks, there are a finite number of ways that ties can be arranged. This is called the “sample space,” and in large networks, this number of possible arrangements can be incredibly large. For a directed network, the number of possible network states is $2^{n(n-1)}$, where n is the number of nodes: this is a (very) large number for even a 38-node graph.¹ Then, for a distribution of graphs, we have to assign a probability to each of this really

¹ In fact, already for $n = 17$, the number of possible directed graphs exceeds what has frequently been quoted as the number of atoms in the observable universe, as $2^{17(17-1)} \approx 10^{80}$ (Wikipedia, [2012](#)).

large number of graphs. Model estimation assigns probabilities so that an observed network is central, not extreme, in terms of the effects being modeled.

4.3 Some Basic Ideas about Statistical Modeling

It is important to understand why we *model* a network. Models represent theories we may have about observed data, and fitting a model permits us to see if our theoretical conception about the data can be validated. A desirable goal of a model is to best represent our observed data – to reproduce the structures we witness in our observed network. In an ERGM, our model represents the combination of structures of which our observed network is composed, permitting inferences about the processes of network tie formation. As noted, when applying an ERGM to observed data we are in fact estimating a model for that data, just as we do when we apply a regression model to non-network data.

Outdated (and unreliable) methods of estimating an ERGM used standard logistic regression procedures. Any comparison with logistic regression breaks down decidedly once network dependencies are introduced because logistic regression assumes independent observations. However, there are some analogies that may be helpful to readers who are new to ERGMs. In a logistic regression, we predict a binary variable from a number of predictor variables, with model parameters (regression coefficients) indicating how important a predictor variable is in that prediction. Similarly, with an ERGM, we predict the presence of a network tie (a binary variable) from several predictor variables (the types of network configurations in which that tie is involved), with model parameters indicating the importance of a configuration to the presence of a tie.

By predicting individual ties in this way, however, the ERGM assigns a probability to each graph in the sample space and hence produces a graph distribution. The following are some terms to understand:

Statistic: An ERGM statistic always relates to a particular configuration. The statistic is typically the count of configurations of different types in the network (e.g., the number of reciprocated ties), although in some cases it may be complex combination of counts or other functions of the configurations.

Parameter: The parameter is the weight applied to the statistic (just as in a logistic regression with predictor variables and regression coefficients). In that sense, it indicates the importance of a statistic, and thereby the importance of a configuration in the network.

An ERGM then assigns a probability to a graph by a (function of a) sum of statistics weighted by parameters, as described at the beginning of [Chapter 2](#). For the moment, it suffices to understand that in

the graph distribution, a positive parameter assigns higher probability to those graphs with many of the relevant configurations. For instance, if a transitive triad parameter is large and positive, then graphs with many transitive triads are more probable (in the graph distribution for that model). If our model is a good explanation for real networks, then the observed graph has more transitivity than a random graph. This fits in nicely with a graph distribution where highly transitive graphs are more probable. However, if a transitive triad parameter is large and negative, then graphs with fewer transitive triads become more probable under our model. Throughout, we are assuming that the data we see are very likely, not unlikely (and so the method of estimation is called “maximum likelihood”).

Simulation: One can formulate an ERGM and then simply select some parameter values to simulate the model to see the type of graphs produced. In Figure 4.2, we learned from a simulation that graphs generated by U|L with 38 nodes and 146 arcs tend to have less than 15 reciprocated arcs. Thus, a simulation tells us what to expect from a particular model with parameter values that we choose. We can study any graph feature of interest in this way, not just reciprocated arcs.

Estimation: However, we also know that for the purposes of understanding reciprocity, the communication network in Figure 4.1, with 44 reciprocated arcs, has significantly more reciprocity. So, for observed data, instead of merely choosing values for the parameters in an ad hoc way, we can systematically search through possible parameter values until we find the right *estimates* so that observed statistics are not extreme, but are central, in the distribution of graph statistics from the resulting ERGM.

Standard error (SE): Each parameter estimate has a SE, which is a measure of the precision, or how certain we are, of the parameter estimate. A small SE indicates greater precision and certainty, whereas a large SE indicates less certainty. We know that a large and positive parameter estimate indicates that more of the associated configurations are likely to be seen in the network. The SE can give us an indication of what to consider as “large.” In this book, we typically use the rule that a significant parameter is one with an absolute value more than twice its SE (see Section 12.5.1 for more detail).

4.4 Homogeneity

When considering reciprocation in the communication network in Figure 4.1b, we are not interested in knowing specifically if there is a reciprocated tie for each pair of actors. Rather, we are interested in the overall tendency for reciprocity in the network. Typically, an ERGM

produces one estimate of reciprocity for *all pairs* of actors across the network. Of course, some individuals may be more inclined than others to reciprocate network ties, but we can consider the estimate as indicative of an average effect across the network. Inherent here is the assumption that local network configurations occur homogeneously across the observed network.² The use of actor attributes is one way to relax homogeneity. Boys and girls in a school may have different levels of average reciprocity, and so we could have different reciprocity parameters for each.

With a basic understanding of the methodological details of ERGMs in hand, we now proceed to show a simple example application of an ERGM to a social network of interest. In doing so, we demonstrate how competing explanations about how network ties arise can be statistically compared within a single model.

² Homogeneity assumptions are not something particular to ERGM. All general linear model techniques assume homogeneity. For instance, in a regression, a case is only identified up to the values of the variables that represent it.

Example Exponential Random Graph Model Analysis

Dean Lusher and Garry Robins

Exponential random graph models (ERGMs) permit us to examine competing theories regarding the formation of network ties, all within a single analysis. This means that a researcher can test one network theoretical concept for tie formation against another (e.g., Does homophily explain this network, or is it reciprocity?) for their network of interest.

This chapter aims to make the ERGM framework clearer through a simple illustrative example. We only use the terminology introduced in the previous chapters, and leave a full description of the parameters and so on to later chapters in Section II. A more detailed analysis of this example is found in [Chapter 13](#).

5.1 Applied ERGM Example: Communication in “The Corporation”

As a practical example of the issues facing social network researchers, we now present the application of ERGMs to the following organizational case study. This example network draws together a number of issues presented so far. The organization of interest is called “The Corporation,” a fictitious name for a real organization in the entertainment industry, which consists of thirty-eight executives. The network under investigation (our “observed network” on which we have collected data) is the “communication network,” which is binary and directed. A tie here represents a response from an actor to a survey item about other executives with whom it was important to communicate to get work completed effectively. The network is depicted in Figure 4.1b. Some basic statistics of the network were presented in Table 4.1.

The Corporation is in competition with other organizations for market dominance. We are interested in understanding the structure of informal communication ties. We also have information on the executives

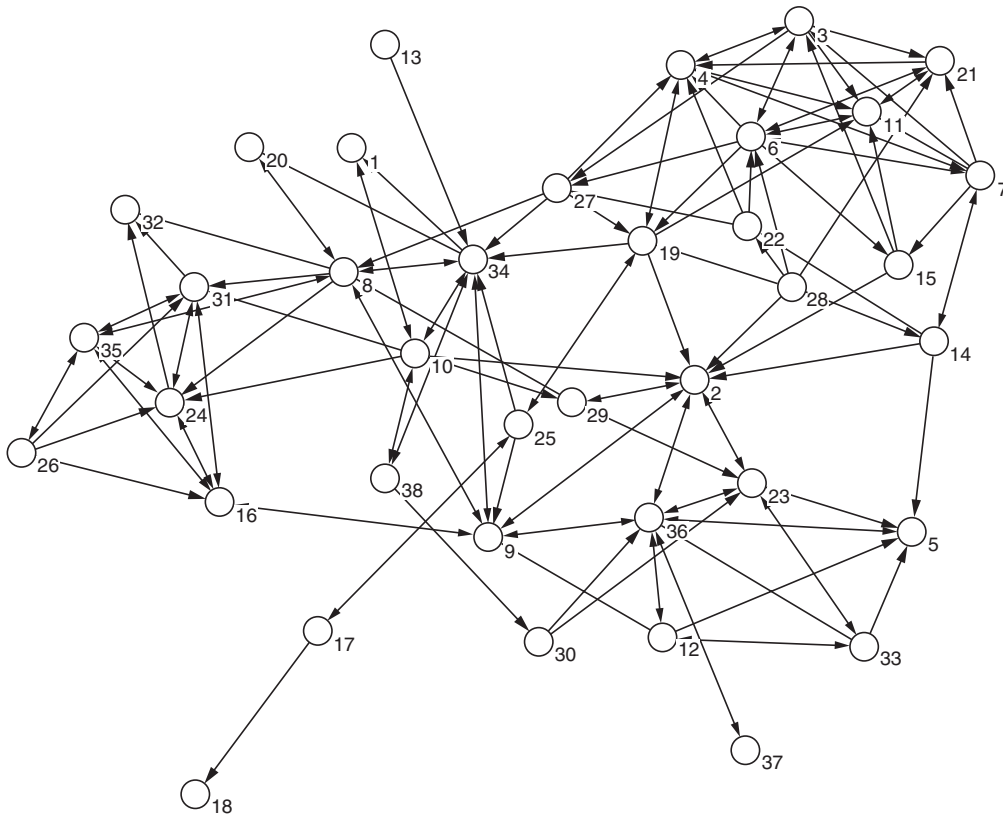


Figure 5.1. Communication network of The Corporation ($n = 38$ actors).

themselves, such as the number of projects they have been involved with, their level of seniority in The Corporation, and their office membership. Finally, we also have information about advice relations within the organization, a binary, directed network of relations with ties pointing toward executives from whom advice is received. The following visualizations display the structure of communication ties within The Corporation in isolation and also in relation to both actor attributes and the covariate network.

This data set can be downloaded in its entirety from MelNet (<http://www.sna.unimelb.edu.au/>) so that readers can fit the models themselves and gain experience in how ERGMs can be applied to real data. The models here are fit with PNet which can be downloaded from the same website.

Network self-organization may be responsible for communication ties. What is evident from Figure 5.1 is the number of triads in the network. With the exception of a few individuals, most employees are within some form of triadic structure. In fact, some regions of the network are very densely clustered. Furthermore, it appears that some nodes (e.g., twenty-four and thirty-four) act as hubs and receive more ties than others in the network.

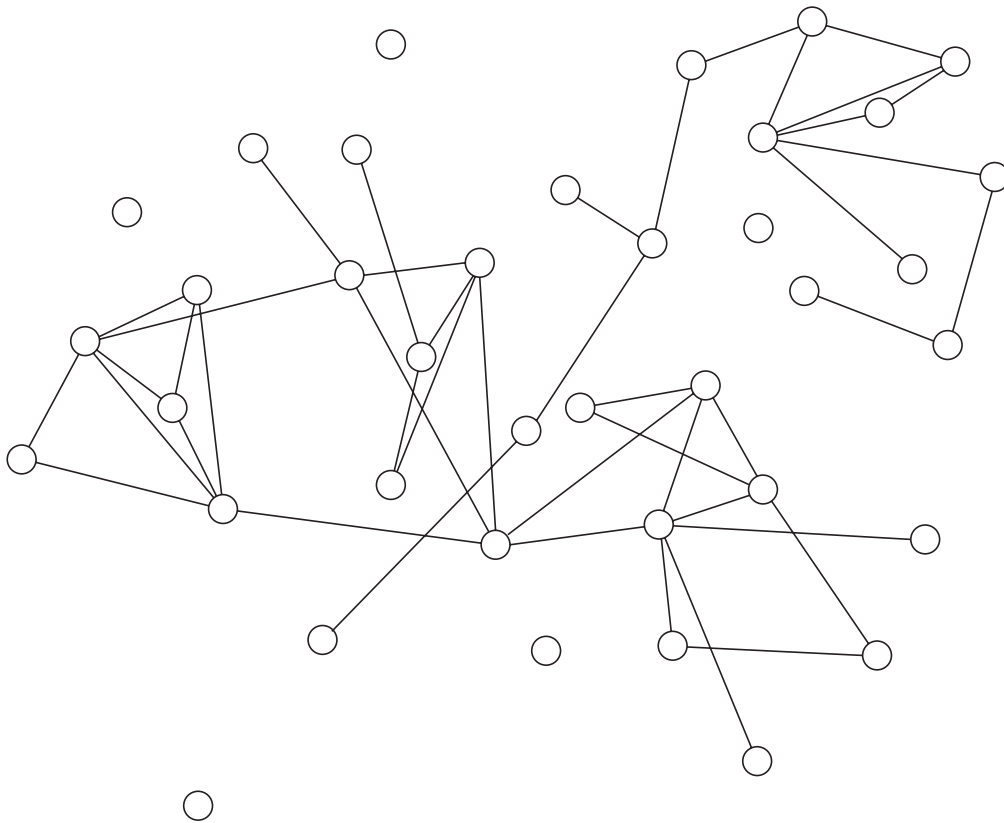


Figure 5.2. Mutual ties only (asymmetric ties removed) in communication network.

Figure 5.2 presents the mutual ties in the network (i.e., undirected ties derived from reciprocated ties in the original data), of which there are many, so it is possible that communication is at least partially explained by reciprocity.

Actor attributes may be important to network structure. For instance, Figure 5.3 depicts node sizes as varying according to the number of projects completed by the executives, with larger nodes indicating more projects. It suggests that there may be a homophily effect based on number of projects (experience).

Figures 5.4 and 5.5 represent seniority of the executives and the branch office of each executive, respectively. It seems that there could be homophily due to seniority, but branch office effects are not so clear from the visualization.

Communication ties may be affected by the presence of advice relations among these executives. Figure 5.6 presents those ties that align (i.e., are entrained) between communication and advice networks. There are twenty-four aligned ties between these networks, although it is difficult to say from the visualization whether this is due to mere chance or whether there is a distinct entrainment effect.

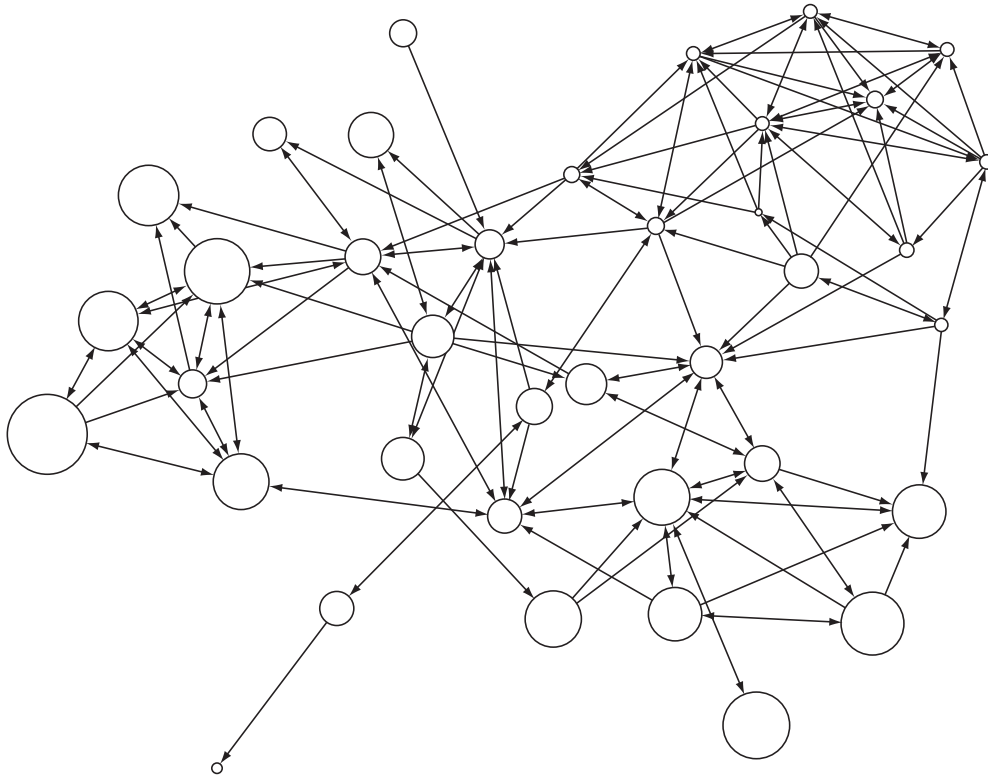


Figure 5.3. Communication network with employee experience represented by size (larger = more experience).

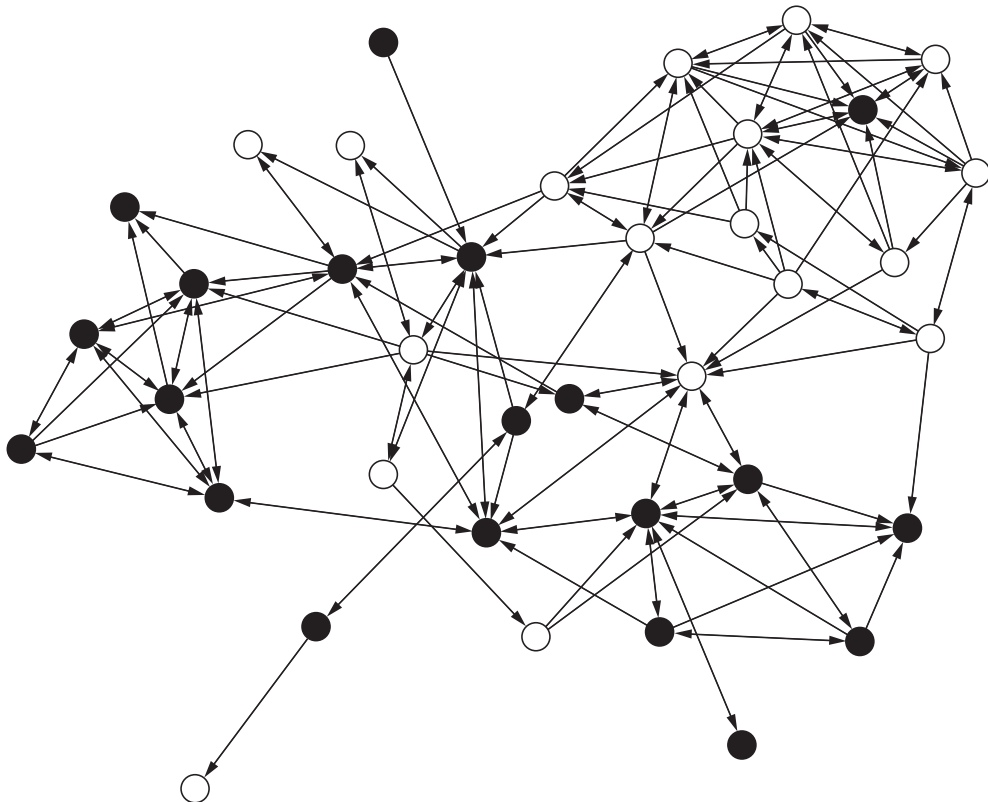


Figure 5.4. Communication network with seniority (black = senior, white = other).

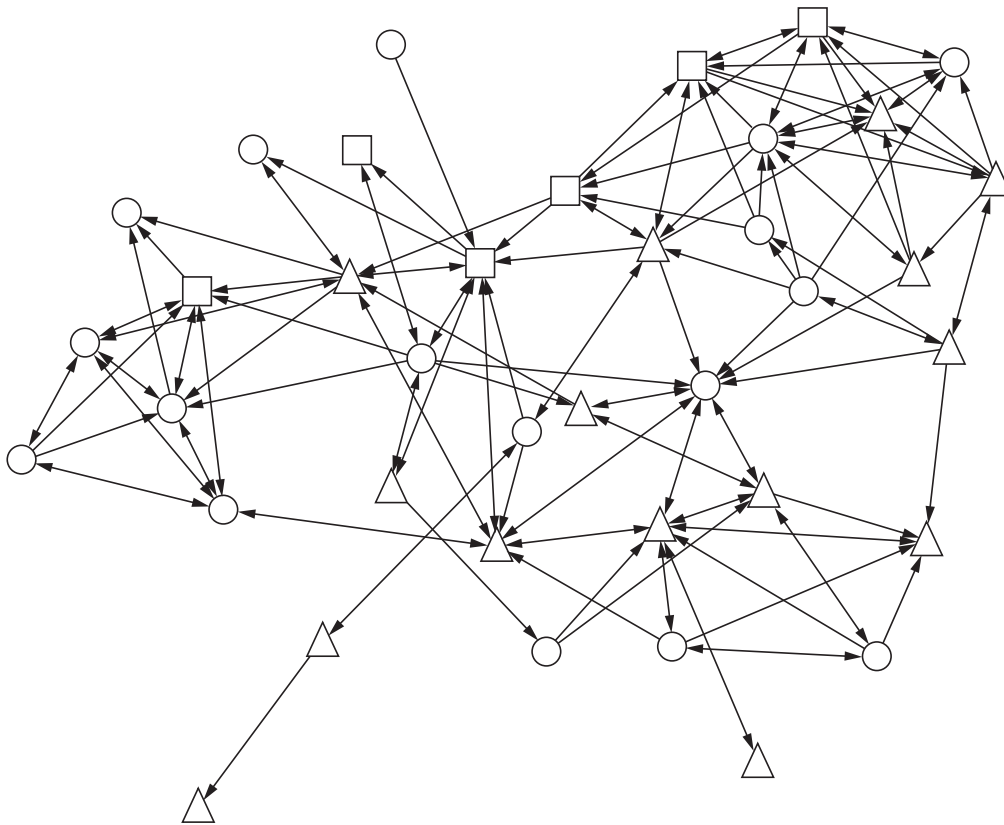


Figure 5.5. Communication network with office membership represented by shape.

So, there are many possible competing explanations for the network structure of communication ties. If we examine only one process at a time, then we may overestimate its worth. With an ERGM, we can infer whether there are independent tendencies for a particular configuration to occur in the network, or whether the presence of that configuration can be more parsimoniously explained by the combined presence of other effects in the model.

5.2 ERGM Model and Interpretation

The model produces parameter estimates that indicate the strength and direction of network patterns. The parameter estimates (and standard errors) for one model of the communication network are presented in [Table 5.1](#). Significant parameters¹ are designated by an asterisk (*). A positive (negative) estimate indicates more (less) of the configuration in the network than expected (given the other effects in the model). We

¹ See Section 12.5.1 for an explanation of significant effects.

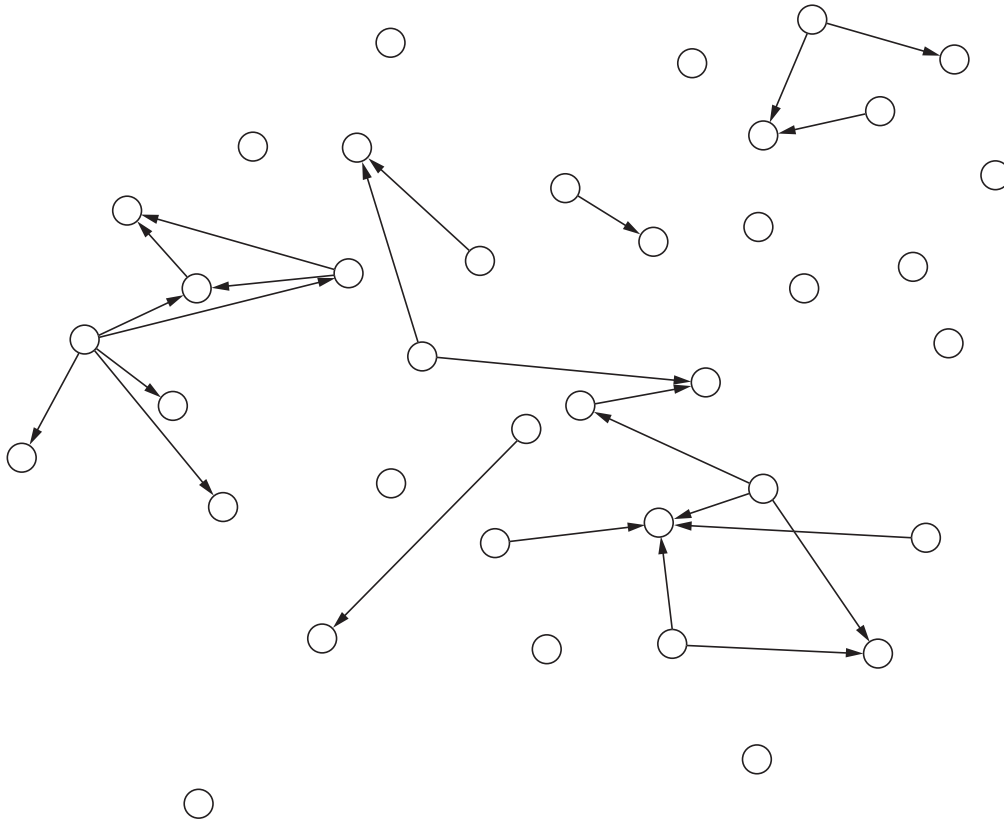


Figure 5.6. Multiplex communication and advice ties (all other ties removed).

include visualizations of each network effect for clarity. The magnitudes of parameter estimates are not directly comparable across different effects because the scaling of the statistics is different. In presenting this model, we have used simplified parameter names based on the general terminology used in this chapter. This model, as well as comparative models for this data, are presented in much more detail in [Chapter 13](#).

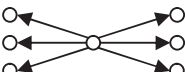
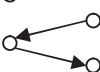
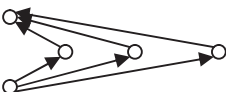
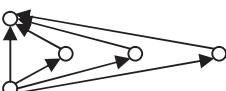
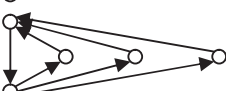
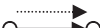
Arc: First, we have a negative *arc* effect. The arc effect is like an intercept effect in a linear regression and can be interpreted as the baseline propensity for the occurrence of ties.²

Reciprocity: The reciprocity estimate is positive and significant. Employees are likely to reciprocate communication.

Popularity and activity (in- and out-degree effects): These effects represent tendencies for centralization in the in- and out-degree distributions, respectively. The in-degree parameter estimate is negative but nonsignificant, suggesting no distinctively popular employees in this company, net of other effects in the model. The out-degree effect is negative and significant, indicating an absence of centralization in network activity,

² The arc parameter is *not* a direct measure of network density.

Table 5.1. ERGM parameter estimates (and standard errors) for communication relations in *The Corporation*

Network effect		Estimate (SE)
Purely structural effects (endogenous)		
Arc		-1.96 (0.73)*
Reciprocity		2.88 (0.46)*
Popularity (in-degree)		-0.27 (0.32)
Activity (out-degree)		-0.34 (0.34)
Simple 2-path ³		-0.06 (0.08)
Multiple 2-paths		-0.06 (0.09)
Transitivity (transitive path closure of multiple 2-paths)		1.22 (0.19)*
Cyclic closure (cyclic closure of multiple 2-paths)		-0.37 (0.17)*
Actor relation effects (exogenous) (black nodes indicates actor with attribute)		
Sender (seniority)		-0.56 (0.29)
Sender (projects)		0.01 (0.02)
Receiver (seniority)		0.08 (0.23)
Receiver (projects)		-0.02 (0.02)
⁴ Homophily (seniority)		0.64 (0.26)*
Heterophily (projects)		-0.08 (0.02)*
Homophily (office)		-0.01 (0.17)
Covariate network (exogenous)		
Advice entrainment (covariate arc)		1.76 (0.30)*

* = parameter estimate is greater than two times the standard error in absolute value, indicating the effect is significant (see Section 12.5.1 for details).

³ The term “simple 2-path” is used to contrast the single 2-path (simple connectivity) with the multiple 2-paths (multiple connectivity) noted in Table 5.1. To be clear, a simple 2-path parameter is also known as a 2-path or mixed 2-star parameter.

⁴ Homophily (and heterophily, its logical opposite) works differently for binary, continuous, and categorical variables. This is taken up in detail in Chapter 8, and explained again in Section III. For binary and categorical variables, we measure homophily – that is the choice of similar others. For continuous variables, we measure the difference between the attributes of actors. As such, a positive parameter estimate would indicate heterophily (i.e., large differences in scores), and a negative estimate would indicate homophily (i.e., small differences in scores).

so that people tend to be rather uniform in the number of choices of communication partners.

Simple 2-path: This parameter is not significant, indicating that we see neither more nor less 2-paths than we might expect given the other effects in the model. That is, there is no evidence that people who send more ties also receive them. If this parameter were significantly positive, for instance, it would suggest that actors who were the most popular were also the most active (i.e., a positive correlation between the in- and out-degree distributions).

Multiple 2-paths: In addition to the single 2-path, it is often desirable to have a multiple 2-paths parameter in the model to indicate the “depth” of local connectivity between pairs of nodes. Notice our contrasting interpretations: the simple 2-path interpretation focuses on the node at the center of the two path (with both an incoming and outgoing tie), whereas the multiple 2-path parameter is best considered as explaining connectivity between pairs of nodes at the end of the paths. Here, this parameter is not significant, indicating that local connectivity is neither stronger nor weaker than we expect given the other effects in the model.

Transitivity: There is a significant and positive effect for transitivity, indicating a tendency for hierarchical path closure in this network. Notice from the diagram that we do not use single triangles to model this effect, but multiple transitive triads based on multiple 2-paths. As is seen later, there are important technical reasons why single triangle parameters are problematic for ERGMs and typically do not result in coherent models.⁵ This reflects an important feature of many empirical social networks: triangles tend to occur together in denser regions of the network. Multiple 2-paths are nested within multiple triangles, so the combination of the two parameter estimates provides an important elaboration of our previous interpretation of the multiple 2-path parameter. When multiple 2-paths occur, they tend to be closed in transitive form (positive transitivity estimate).

Cyclic closure: There is a significant and negative effect for cyclic closure, indicating a lack of nonhierarchical (or generalized exchange) network closure in this network. Again, the cyclic closure form is associated with multiple 2-paths, so we can further elaborate our interpretation of the parameters together. When multiple 2-paths occur, they tend to be closed in transitive form and tend not to be closed in cyclic form; once these two closure processes are taken into account, there are no other evident tendencies for multiple 2-paths to be present or absent. In other words, local connectivity is explained by tendencies for transitive closure and tendencies against cyclic closure.

⁵ Technically, models with single triangle parameters very often do not converge. This is discussed in [Chapter 6](#).

Sender effects: The sender effect measures the degree to which actors with a specific attribute send more ties compared to others in the network. The negative but not quite significant sender effect for the binary attribute seniority means that managers who are not senior (i.e., those who have a score of zero) have a tendency to send ties. In contrast, the number of projects was measured as a continuous attribute variable. However, there were no sender effects for number of projects completed, so there is no evidence here that experience in projects affects tendency to communicate. Finally, note that there is no sender effect (or receiver effect) for the variable “office” because it is a categorical variable.

Receiver effects: The receiver effect measures the degree to which actors with a specific attribute have the propensity to receive ties. There were no significant receiver effects for the variables of seniority and number of projects.

Homophily: There is a significant and positive estimate for homophily for seniority and the number of projects. Actors tend to communicate with others of the same seniority and with others of similar experience in terms of the number of projects. We note that the effect is negative for projects, here measured as the (absolute) difference, and thus a negative value indicates small differences. The estimate for the categorical variable office homophily was not significant.

Covariate advice network: The covariate advice network has been treated as exogenous to the communication network, and so is fixed in the modeling process. Theoretically, this means that we assume advice may have an effect on communication (the network we are modeling) but that communication has no effect on advice (for an extended discussion on this, see [Chapter 21](#)). More to the point, we seek to understand how advice ties might explain communication ties but not the other way around. The significant and positive parameter of the covariate network indicates that advice and communication ties co-occur, or are entrained.

5.2.1 Multiple Explanations for Network Structure

First and foremost, when examining the estimates in [Table 5.1](#), we highlight the fact that there are significant effects for purely structural, actor-relation, *and* covariate network effects. That is, each subtype of network effects has an independent explanatory capacity for the presence of ties in the communication network. This emphasizes a number of points. If we were to believe that social network ties were not dependent on the individual characteristics of actors, we would miss important information, specifically, how communication occurs around similarity on a number of characteristics in The Corporation. Furthermore, exogenous relational structures (i.e., advice) also have a unique impact on communication ties. The purely structural effects by themselves do not wipe out the

actor-relation effects, and thus this highlights the importance of examining the qualities of actors in the network. Conversely, the inclusion of attributes does not erase the purely structural effects found for the network, indicating that attributes alone are insufficient for explaining the formation of network ties.

We hope that the example in this chapter gives an early illustration of how an ERGM may be applied to understand network structure. With this motivation behind us, we now turn to more precise methodological details about how ERGMs are formulated and implemented in Section II.

Section II

Methods

6

Exponential Random Graph Model Fundamentals

Johan Koskinen and Galina Daraganova

6.1 Chapter Outline

This chapter provides a more detailed description of exponential random graph models and aims at answering the following questions:

- What do the different notions of independence and interdependence imply for the modeling of tie-variables?
- What are exponential random graph models? What can they tell us?
- Why and when should we use exponential random graph models?
- What different model specifications are available? How may these be interpreted?

After presenting some necessary notation, we revisit the concept of statistical independence in order to move to an understanding of interdependence. A network approach implies some level of dependence among the observations. We then explain the exponential random graph model (ERGM) framework akin to more familiar generalized linear models, emphasizing that we now have dependence, not independence, of observations. Next, the exact nature of these departures from independence is explained, and the implications for model specifications are presented. We begin by explaining individual ties because this illustrates the snug fit between individual ties, endogenous dependencies, and the model expressed in terms of the entire graph.

6.2 Network Tie-Variables

For a given undirected graph, a set of n nodes is represented as $N = \{1, \dots, n\}$, and $i \in N$ means that “ i belongs to the set N .” This set is assumed to be fixed and predetermined. Let J be the set of all possible relational

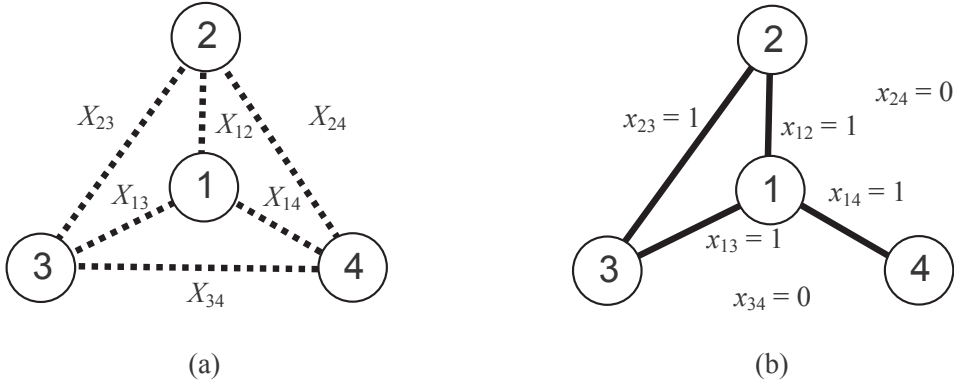


Figure 6.1. (a) Network variables of X and (b) a realization x for network on four vertices.

ties for the node set N , $J = \{(i, j) : i, j \in N, i \neq j\}$. This set excludes pairs (i, i) because self-ties are disallowed. The number of elements¹ in J is $\binom{n}{2} = \frac{n(n-1)}{2}$. For any observed network, some edges in J may be present and some absent. For a stochastic model, the set of ties E that are present is a random subset of J . If $(i, j) \in E$, then the edge between i and j is present. For any element $(i, j) \in J$, a random variable X_{ij} can be defined where $X_{ij} = 1$ if $(i, j) \in E$ and $X_{ij} = 0$ if $(i, j) \notin E$. This description is for an undirected graph: the same ideas applies to a directed graph with some slight variation in details (e.g., the number of elements in J is then $n(n-1)$).

These variables are called “tie-variables.” The tie-variables can be collected in a stochastic adjacency matrix $X = [X_{ij}]$, where the entry in row i and column j pertains to the tie from i to j . The space of all possible adjacency matrices is denoted by $\mathbf{X}$. A realization of the stochastic adjacency matrix X is denoted by $x = [x_{ij}]$, a realization being a particular instance of the stochastic adjacency matrix – here, a matrix of actual zeros and ones rather than variables. Any observed network, then, is a realization.

The collection of network tie-variables and a particular realization for a network on four nodes is represented in Figure 6.1. The left panel of Figure 6.1 schematically represents the collection of six tie-variables that constitute a random graph on four vertices. The dotted line between nodes 2 and 3, for example, represents the tie-variable X_{23} . One realization of the random graph is depicted in the right-hand panel. Here, the tie between 2 and 3 is present, indicated by a solid line, but the tie between

¹ In general, the number of ways in which one may choose k -subsets out of n elements is given by

$$\binom{n}{k} = \frac{n(n-1) \cdots (2)(1)}{k(k-1) \cdots (2)(1)(n-k)(n-k-1) \cdots (2)(1)} = \frac{n(n-1) \cdots (n-k+2)(n-k+1)}{k(k-1) \cdots (2)(1)}$$

Table 6.1. *Two independent tie-variables*

		Tie (i, j)	
		Prob absent: .5	Prob present: .5
Tie (h, m)	Prob absent: .5	Prob $(h, m), (i, j) \notin E$: .25	Prob $(h, m) \notin E, (i, j) \in E$: .25
	Prob present: .5	Prob $(h, m) \in E, (i, j) \notin E$: .25	Prob $(h, m), (i, j) \in E$: .25

vertices 2 and 4 is absent – the realizations of the corresponding tie-variables are $x_{23} = 1$ and $x_{24} = 0$, respectively.

The main focus is on modeling a process of network tie formation. As mentioned in [Chapter 3](#), the presence of a tie between any two individuals may be explained by a combination of actor attribute variables and by patterns of ties among other members of the network. In other words, the likelihood of a tie may not only be a function of individual characteristics of actors who share the tie, but also a function of presence or absence of other network ties in the network. This possible dependence between tie-variables creates the main difficulty in formulating a stochastic model for a network.

6.3 Notion of Independence

To understand statistical dependence, let us begin with the idea of independence (see, e.g., Newbold, Carlson, and Thorne (2007) for an introduction to statistical concepts). The assumed independence of two tie-variables, X_{ij} and X_{hm} , may be compared to coin tosses. If we assume that the ties (h, m) and (i, j) are equally likely to be present and absent, we may think of a model for these ties as two coins, coins I and II, so that when we flip them separately, each has the probability of .5 of coming up heads and tails. If a coin is heads, we deem the corresponding tie to be present, and if it is tails, we deem the tie absent.

If the coinflips are independent, this means that the probability that they will both come up heads is just the product of the individual probabilities of coming up heads, as in [Table 6.1](#). This is a statistical model for a network where the density is expected to be .5.

Suppose instead that the tie-variables behaved as in [Table 6.2](#), where multiplying the probabilities of “heads” does not produce the joint probabilities. In this case, the variables are not independent. The marginal probabilities are the same as in [Table 6.1](#); the probability of (h, m) being present is still .5 ($= .1 + .4$).

An equivalent definition of independence of two variables, X_{ij} and X_{hm} , is that the information that $X_{hm} = x_{hm}$ does not alter the probability that $X_{ij} = x_{ij}$. In other words, information regarding the realization of one variable does not help us predict the value of the other variable – the

Table 6.2. *Two dependent tie-variables*

		Tie (i, j)	
		Prob absent: .5	Prob present: .5
Tie (h, m)	Prob absent: .5	Prob $(h, m), (i, j) \notin E$: .4	Prob $(h, m) \notin E, (i, j) \in E$: .1
	Prob present: .5	Prob $(h, m) \in E, (i, j) \notin E$: .1	Prob $(h, m), (i, j) \in E$: .4

unconditional (or marginal) probability is still our best guess for the second variable.

Most standard statistical approaches such as logistic regression require independence, as in Table 6.1, whereas for networks we generally expect dependence of some sort, as in Table 6.2. For instance, if we know that Mary talks to John and that John talks to Peter, we expect that the chance of Mary talking to Peter is greater than just the marginal probability of two randomly selected people in the whole world talking to each other (which is very small indeed). In a directed network, it does not seem plausible that a coin flip for X_{ij} is independent of a coin flip for X_{ji} because these variables relate to the same dyad. Thus, if John invites Jack to dinner, the chances of Jack inviting John to dinner are likely to increase. Furthermore, for a directed or undirected graph, it does not seem plausible that a coin flip for X_{ij} is independent of a coin flip for X_{ih} because both tie-variables pertain to the same actor, i (this is the reasoning underlying the Markov dependence assumption explained later in this chapter).

In these cases, there is dependence among tie-variables, and the presence of some ties affects the probability that other ties may be observed. Ignoring dependence among observations has major consequences for the statistical conclusions we draw, and this is well documented in related fields (Snijders & Bosker, 1999). Exponential random graph models have the explicit aim to relax the assumption of independence between network tie-variables and to incorporate possible dependencies among tie-variables. Conditioning – “knowing that Mary knows John” – is a central part of model formulation, interpretation, and simulation. ERGMs do not aim at predicting the probability of a tie in a social vacuum, as in the case of a logistic regression, but rather the “conditional probability” of a tie, given what is observed in the rest of the network.

6.4 ERGMs from Generalized Linear Model Perspective

Although ERGMs have some similarities to generalized linear models (GLMs), especially standard log-linear models and logistic regressions,

an ERGM *does not* (except in trivial cases) reduce to logistic regression. Yet, because some fundamental concepts are common to GLMs and ERGMs, it is instructive to illustrate these in the familiar language of logistic regression. This also serves to emphasize how the main departure of ERGMs from logistic regression, namely, the assumptions of dependence between observations, play out. Although logistic regression assumes independence of observations as in Table 6.1, ERGMs do not make this assumption, rather the opposite.

Suppose that we are primarily interested in explaining observed ties as a function of a collection of covariates, or predictor variables. The covariates for the tie-variable X_{ij} could, for example, relate to the individual characteristics of the two actors i and j , such as the difference in age between i and j , and a variable indicating whether i and j have the same gender. Denote these dyadic covariates by $w_{ij,1}, w_{ij,2}, \dots, w_{ij,p}$, for p covariates. For a GLM, we would try to find a function η of w and unknown parameters $\theta_1, \theta_2, \dots, \theta_p$ that best describe the expected value $E(X_{ij}) = \eta(w, \theta)$ (the probability that $X_{ij} = 1$). For dichotomous response variables such as X_{ij} , a logistic regression estimates a set θ of unknown parameters $\theta_1, \theta_2, \dots, \theta_p$ (logistic regression coefficients) that best predict the probability that the tie is present. The logistic regression function is

$$\begin{aligned} \Pr(X_{ij} = 1|\theta) &= \eta(w, \theta) \\ &= \frac{\exp\{\theta_1 w_{ij,1} + \theta_2 w_{ij,2} + \dots + \theta_p w_{ij,p}\}}{1 + \exp\{\theta_1 w_{ij,1} + \theta_2 w_{ij,2} + \dots + \theta_p w_{ij,p}\}}. \end{aligned}$$

If a covariate, say, $w_{ij,2}$, indicated whether i and j were of the same gender, a positive value of the corresponding parameter θ_2 indicates a higher probability of a tie between people of the same gender. It is usually easier to interpret the model in terms of the logit or log-odds, which is the natural logarithm of $\Pr(X_{ij} = 1|\theta) / \Pr(X_{ij} = 0|\theta)$:

$$\begin{aligned} \text{logit } \Pr(X_{ij} = 1|\theta) &= \log \frac{\Pr(X_{ij} = 1|\theta)}{\Pr(X_{ij} = 0|\theta)} = \theta_1 w_{ij,1} + \theta_2 w_{ij,2} + \dots \\ &\quad + \theta_p w_{ij,p}. \end{aligned}$$

Anyone familiar with linear and/or logistic regression will be comfortable with the expression on the right-hand side. The parameters (θ) weight the relative importance of their corresponding predictors (w) for the probability of a tie. Positive parameters correspond to effects that increase the probability of a tie, whereas negative parameters relate to effects that decrease the probability of a tie.

The difference in the log-odds for two pairs (i, j) and (h, m) , the covariates of whom only differ in that i and j are of the same gender ($w_{ij,2} = 1$)

and h and m are of different gender ($w_{hm,2} = 0$), is

$$\frac{\text{logit Pr}(X_{ij} = 1|\theta)}{\text{logit Pr}(X_{hm} = 1|\theta)} = \theta_1(w_{ij,1} - w_{hm,1}) + \theta_2(w_{ij,2} - w_{hm,2}) + \cdots \\ + \theta_p(w_{ij,p} - w_{hm,p}) = \theta_2.$$

This ratio (of being of the same gender) is the well-known odds ratio. The larger the value of θ_2 , the greater the probability of a tie for same-gender pairs as compared to different-gender pairs, everything else being equal. We can think of θ_2 as relating to the *change* in going from a situation of a different gender to a same-gender pair with everything else the same.

For ERGMs, in addition to the exogenous covariates used in logistic regression such as the w variables, we include as covariates counts of “network configurations” in the linear predictor. Configurations were introduced in [Chapter 3](#), and examples include edges, 2-stars, and triangles (see Section 3.1.2). In addition, we provide details of other possible configurations later in this chapter. The interpretation of the parameters corresponding to these configurations is similar to those of exogenous covariates; for example, a positive parameter corresponding to the number of triangles means that a tie is more likely to occur if it closes a 2-path than if it does not. In the example of the triangle, as the reader will notice, whether a tie closes a 2-path depends on whether the other two ties of the triangle are present.

Consequently, the second departure from logistic regression is that we have to formulate the model for each tie-variable conditional on the rest of the graph: that is, in predicting a tie X_{ij} , we need to take into account the other ties that might be present. In other words, ERGMs predict the probability for X_{ij} , conditional on all other ties observed in the network (which we denote as X_{-ij}). This conditional probability is written as $\text{Pr}(X_{ij} = 1|X_{-ij} = x_{-ij}, \theta)$. Leaving aside the dyadic covariates (the w variables previously mentioned), and concentrating only on the configuration counts as predictors, the (conditional) logit then becomes

$$\log \frac{\text{Pr}(X_{ij} = 1|X_{-ij} = x_{-ij}, \theta)}{\text{Pr}(X_{ij} = 0|X_{-ij} = x_{-ij}, \theta)} = \theta_1 \delta_{ij,1}^+(x) + \theta_2 \delta_{ij,2}^+(x) + \cdots \\ + \theta_p \delta_{ij,p}^+(x). \quad (6.1)$$

The functions $\delta_{ij,k}^+(x)$ are called the “change statistics” for the k th configuration. They are not just counts of the configurations in the graph (e.g., the number of triangles) but the change in going from a graph for which $X_{-ij} = x_{-ij}$ and $X_{ij} = 0$ to a graph for which $X_{-ij} = x_{-ij}$ and $X_{ij} = 1$. For example, if one covariate is the number of edges, then adding the edge (i, j) to $X_{-ij} = x_{-ij}$ will result in an increase in the number of edges by one, say, $\delta_{ij, \text{edge}}^+(x) = 1$. Adding the edge (i, j) to $X_{-ij} = x_{-ij}$ when $x_{ik} = x_{kj} = 1$ will result in an increase in the number of triangles by

(at least) one because this would create a new triangle $x_{ij} = x_{ik} = x_{kj} = 1$. If the parameter corresponding to the number of triangles is positive, then the fact that the triangle count increases would contribute positively to the probability for $X_{ij} = 1$.

Note here the important fact that we need to know the rest of the graph $X_{-ij} = x_{-ij}$ in order to calculate the $\delta_{ij,k}^+(x)$ and the conditional logits. This is a direct consequence of the assumption that ties may be interdependent – the probability of a tie depends on whether other ties are present. The probabilities (or probability distributions) presented in this chapter may be interpreted conditionally – an ERGM prescribes how likely it is to add or delete a tie for a pair of actors given everything else. These probabilities are based on the weighted contributions of changes in configurations that adding or deleting the tie in question would yield.

Why are the predictors the change statistics of configurations rather than the raw count of configurations? In Equation (6.1), we have an expression for the log-odds for the presence of a tie on X_{ij} compared to the absence of a tie on X_{ij} . In that case, the correct predictor is the change from the graph, when $x_{ij} = 0$, to the graph, when $x_{ij} = 1$.

There is an equivalent form of the model as a probability expression for all tie-variables simultaneously, where the predictors are then the counts of configurations. This is known as the joint form of the model:

$$\Pr(X = x|\theta) \equiv P_\theta(x) = \frac{1}{\kappa(\theta)} \exp\{\theta_1 z_1(x) + \theta_2 z_2(x) + \dots + \theta_p z_p(x)\}. \quad (6.2)$$

Equation (6.2) is the general form of the ERGM that we stick to throughout the book. The functions $z_k(x)$ are counts of configurations in the graph x , such that the corresponding change statistic for $z_k(x)$ would be $\delta_{ij,k}^+(x) = z_k(\Delta_{ij}^+ x) - z_k(\Delta_{ij}^- x)$, where $\Delta_{ij}^+ x$ ($\Delta_{ij}^- x$) denotes a matrix x for which x_{ij} is constrained to be equal to one (zero). The parameters weight the relative importance of their respective configurations, and the normalizing term $\kappa(\theta) = \sum_{y \in X} \exp\{\theta_1 z_1(y) + \theta_2 z_2(y) + \dots + \theta_p z_p(y)\}$ ensures that the sum of the probability mass function, $P_\theta(x)$, over all graphs is one.

Equation (6.2) describes a probability distribution for all graphs with n nodes. Let us suppose that we have only one configuration represented in a model for the network – the number of edges. Then there will be a parameter θ_1 for edges and a statistic $z_1(x)$ that is simply the count L of the number of edges in the graph x . So, for any and every graph x with n nodes, Equation (6.2) with a given edge parameter θ_1 will assign a probability to x based on the number of edges. We can then think of a graph from this probability distribution as a random graph, and due to the form of Equation (6.2), we term it an “exponential (family) random graph distribution.” Because Equation (6.2) is based on certain network

configurations, we can think of graphs in this distribution as built up by the presence and absence of those particular configurations, combining together in ways represented by the parameter values to create the total graph structure.

As the ERGM gives us a distribution of graphs over $\mathbf{X}$, the model also implies a distribution of statistics. This offers a convenient way of studying various properties of a model through inspecting the various implied distributions of statistics (as is done in [Chapter 4](#); the use of simulated distributions of statistics is described in more detail in [Chapter 12](#) (simulation, estimation, and goodness of fit) and further illustrated in [Chapter 13](#)). As an example, we frequently make use of the expected values $E_{x|\theta}\{z(x)\} = \sum_{x \in \mathbf{X}} z(x)P_\theta(x)$ of these implied distributions.

Of course, this description is still quite abstract. To obtain a particular model, we first need to decide which configurations are relevant. We are guided here by hypotheses about possible dependencies among tie-variables.

6.5 Possible Forms of Dependence

We now introduce the different forms of dependence among tie-variables that have been proposed in the literature and briefly note the different types of models that result. With this quick summary of dependence in place, we go on to describe each form of the model in more detail. A more detailed discussion of dependence is provided in [Chapter 7](#).

6.5.1 Bernoulli Assumption

The simplest form of dependence is represented by the Bernoulli graph (e.g., Bollobás, 1985; Frank, 1981; Frank & Strauss, 1986; Karońsky, 1982) in which the tie-variables are assumed to be independent. The Bernoulli graph is called “homogeneous” if the tie-variables are independent and identically distributed Bernoulli variables (in other words, just like a coinflip, although perhaps biased with unequal probabilities for heads and tails). The Bernoulli dependence assumption defines the class of Bernoulli graphs, where the (log-) probability of a graph is proportional to the weighted sum of number of edges. This assumption is quite unrealistic for social networks but can serve as a baseline for comparison.

6.5.2 Dyad-Independent Assumption

For directed graphs, an intuitive form of dependence is to allow for the tie *from* person i to person j to be dependent of the tie *to* person i from person j . Hence, the model is no longer a model for the ties of the network

but for pairs of ties of the network, and the pairs of tie-variables are called “dyads.”

6.5.3 Markov Dependence Assumption

Frank and Strauss (1986) proposed a “Markov dependence assumption,” the simplest assumption that goes beyond a dyad. Two tie-variables are assumed to be independent unless they share a node. If instead of considering the edges of a graph as connecting nodes, we think of the nodes of the graph as connecting the edges, the Markov dependence assumption suggests itself. Because node i connects the possible edges (i, j) and (i, h) , we say that the tie-variables corresponding to (i, j) and (i, h) are *dependent* conditional on the rest of the graph.

This assumption accounts for the fact that whether Mary talks to John may depend on whether Mary talks to Peter because both ties pertain to Mary. In addition, the probability that John talks to Peter may be affected by whether both John and Peter talk to Mary (a possible tie between John and Peter is conditionally dependent on ties between Mary and John and between Mary and Peter under the Markov assumption – note the triangle!). The Markov dependence assumption leads to the class of Markov random graphs, where the (log-) probability of a graph is proportional to the weighted sum of counts of different structural features such as edges, stars, and triangles. How these features are derived is summarized in Chapter 7, and the structural features themselves and how they may be interpreted is dealt with in the next section.

6.5.4 Realization-Dependent Models

Pattison and Robins (2002) suggested that in some circumstances, two tie-variables, X_{ij} and X_{hm} , may be conditionally dependent given the presence of other network tie-variables even when they do not share a node. This is a “partial conditional independence assumption” that is a generalization of the more familiar notion of conditional independence, which refers to the situation of statistical independence of two variables given the state of a third variable (Dawid, 1979, 1980). In the case of partial conditional independence for tie-variables, two tie-variables are statistically independent if and only if a third tie-variable is in a particular state.²

An example of a partial conditional independence assumption is the “social circuit dependence assumption,” where two tie-variables, X_{ij} and

² This approach is similar in form to that used by Baddeley and Möller (1989) for spatial models, which is why Pattison and Robins (2002) adopted the same term referring to these models as “realization-dependent models.”

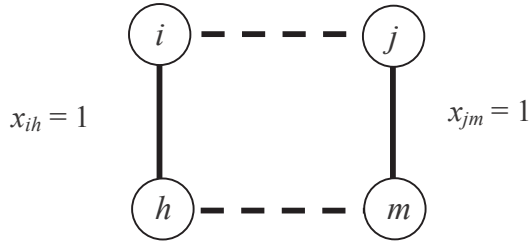


Figure 6.2. Social circuit dependence.

X_{hm} , that do not share a vertex are conditionally dependent if ties exist between i and h and between j and m . In this case, if the two-variables are observed (i.e., $x_{ih}x_{jm} = 1$), then a 4-cycle is created as in Figure 6.2. This dependence may be typified in collaborative ties, when i works with h and j works with m . Then the presence of collaboration between i and j is likely to affect whether h and m also collaborate.

This assumption in combination with the Markov dependence assumption gives rise to a set of additional configurations in the network model, including sets of 2-paths with common starting and ending nodes, and sets of triangles with a common base.

6.6 Different Classes of Model Specifications

A dependence assumption constrains the types of configurations among network tie-variables in the model. This point is argued more fully in Chapter 7, which presents technical details of dependence at greater length. For the moment, it suffices to know that a particular dependence assumption implies that the conditional probability of a tie is affected by certain network configurations. In other words, based on the nature of the dependence, a tie is more (or less) probable if it features in some of these configurations. Counts of these subgraphs in the observed graph then become the statistics in the model that permit us to move beyond logistic regression and take dependence into account. Different model specifications involve different combinations of statistics. We now describe a number of commonly used models based on different dependence assumptions, although this is not a complete list of possible models.

6.6.1 Bernoulli Model

Undirected Graphs. The simplest ERGM is the Bernoulli model. As noted previously, for the homogeneous Bernoulli model, each possible tie can be modeled as the independent flip of a (p -) coin. The interpretation is that for each possible tie, we flip a (p -) coin, where the probability of

heads is p : if this coin comes up heads, we deem a tie present; otherwise, it is absent. The conditional probability is then

$$\Pr(X_{ij} = 1 | X_{-ij} = x_{-ij}, \theta) = \Pr(X_{ij} = 1 | \theta) = \frac{e^\theta}{1 + e^\theta}.$$

The logit of the probability of a tie is given simply by θ , the edge parameter. The joint ERGM probability mass function for the adjacency matrix according to the general formula (Equation (6.2)) is

$$P_\theta(x) = \frac{1}{\kappa(\theta)} \exp\{\theta_L L(x)\}. \quad (6.3)$$

The parameter θ_L is called the “edge parameter,” and the corresponding statistic for this is the number of edges $L(x) = \sum_{i < j} x_{ij}$. There is only one network configuration relevant to this model: the single edge.

Directed Graphs. The homogeneous Bernoulli model for directed graphs follows the same principles as for undirected graphs with the difference that there are twice as many tie-variables.

6.6.2 Dyadic Independence Models

An obvious limitation of the Bernoulli model for directed graphs is that it does not allow for tendencies toward reciprocation, a well-known tendency in human social relations. This prompted the development of the dyad-independent (or p_1) class of exponential family distributions by Holland and Leinhardt (1981) and Fienberg and Wasserman (1981). For two actors $i, j \in N$, the Bernoulli model assumes that the probability that $X_{ij} = x_{ij}$ and $X_{ji} = x_{ji}$ is the product of the two marginal probabilities as in Table 6.1, akin to two independent coin tosses (with biased coins). The dyad-independent model, in contrast, allows for these variables to be dependent as in Table 6.2. Nevertheless, the dependence assumption is limited: the model assumes that dyads are independent of each other, so that the probability of $X_{ij} = 1$ depends only on x_{ji} . Hence, the tie-variables (X_{ij}, X_{ji}) may be treated as independent observations for each pair (i, j) of nodes. Because of the simple dependence assumption, a dyad-independent model may still be fitted using standard statistical techniques (more specifically, multinomial logistic regression).

The simplest form of a dyad-independent model is what Snijders (2002) called the “reciprocity p^* model”:

$$P_\theta(x) = \frac{1}{\kappa(\theta)} \exp\{\theta_L L(x) + \theta_M M(x)\},$$

where $L(x)$ is now the number of arcs and $M(x) = \sum_{i < j} x_{ij}x_{ji}$ is the number of reciprocated (or mutual) arcs in the graph. Here, there are

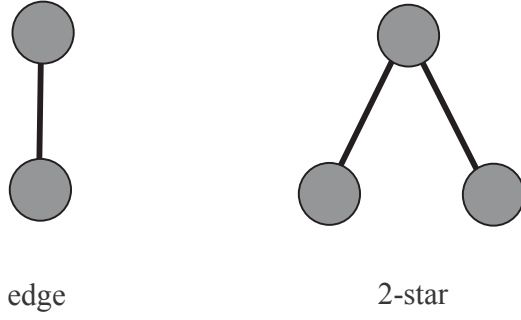


Figure 6.3. Configurations in 2-star model.

two configurations relevant to the model: arcs and reciprocated arcs. For each dyad, there are four possible outcomes for which the associated probabilities are easily seen to be

$$\Pr(X_{ij} = x_{ij}, X_{ji} = x_{ji} | \theta) = \begin{cases} \frac{1}{\kappa_{ij}(\theta)} & x_{ij} = x_{ji} = 0 \\ \frac{1}{\kappa_{ij}(\theta)} \exp\{\theta_L\} & x_{ij} = 1, x_{ji} = 0 \\ \frac{1}{\kappa_{ij}(\theta)} \exp\{\theta_L\} & x_{ij} = 0, x_{ji} = 1 \\ \frac{1}{\kappa_{ij}(\theta)} \exp\{2\theta_L + \theta_M\} & x_{ij} = x_{ji} = 1 \end{cases},$$

where $\kappa_{ij}(\theta) = 1 + 2e^{\theta_L} + e^{2\theta_L + \theta_M}$, obtained from adding the probabilities of the four outcomes.

In the original p_1 class of models, the parameter θ_L was assumed to be a function of sender and receiver effects to reflect an actor's propensity to establish ties to others and to receive ties. In subsequent elaborations of the p_1 , Bayesian smoothing has been employed (Wong, 1987), and more recently, more complicated dependencies have been introduced through random effects in the p_2 model (van Duijn, Snijders, & Zijlstra, 2004).

Although reciprocity is typically not the only type of interdependency encountered in social network data, it is an integral part of most realistic models for directed networks.

6.6.3 Markov Model

Nondirected Markov Random Graph Models. The Markov dependence assumption gives rise to a model whose statistics in Equation (6.2) are the number of edges; the numbers of k -stars $S_k(x) = \sum_{i \in N} \binom{x_{i+}}{k}$ of different sizes ($2 \leq k \leq (n-1)$), where $x_{i+} = \sum_j x_{ij}$ (see Figures 6.3 and 6.4 for depictions of 2- and 3-stars, respectively); and the number of triangles $T(x) = \sum_{i < j < k} x_{ij} x_{jk} x_{ki}$. The Markov dependence assumption supposes that two tie-variables are conditionally dependent if they share a node.

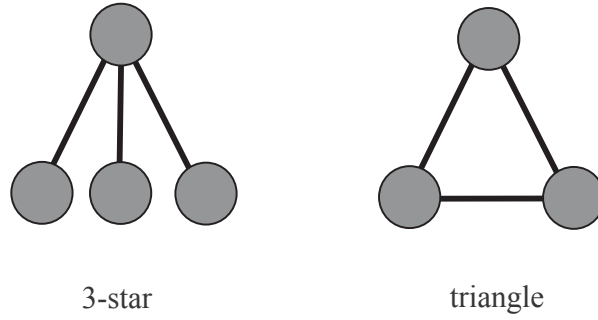


Figure 6.4. Configurations with three edges: 3-star and triangle.

What this means is that any configuration in the model must have ties that all share nodes with each other. A little thought will show that the only way that this can occur is if a configuration is a single edge, if all ties in the configuration share the one node (i.e., a star of different sizes, depending on how many ties there are), or if there are three ties with three nodes (a triangle). (This argument is presented in more detail in [Chapter 7](#).)

The full Markov model then becomes

$$\Pr(X = x) = \frac{1}{\kappa} \exp(\theta_L L(x) + \theta_{S_2} S_2(x) + \theta_{S_3} S_3(x) + \cdots + \theta_{S_{n-1}} S_{n-1}(x) + \theta_T T(x)). \quad (6.4)$$

Subsets of models nested in the full Markov model are obtained by setting some parameters to be zero. Recall (see Figure 3.2 in Section 3.1.5) that the higher-order configurations contain lower-order configurations, so that the statistics are nested. The count of 2-stars ([Figure 6.3](#)) is not just a count of nodes with degree 2: a 3-star contains three 2-stars, as does a triangle ([Figure 6.4](#)). Similarly, a triangle contains three edges, and so on. Thus, statistically, the higher-order configurations represent statistical interactions involving lower-order configurations. This has the effect of permitting inferences about, for instance, the relevance of 2-star configurations given the number of edges in the graph, or the importance of triangles given the number of 2-stars. We want to make inferences about whether there is a distinct process of network ties forming into triangles, over and above the chance accumulation of edges and 2-stars.

Setting different statistics to zero gives rise to different model specifications. As explained later in this chapter, it turns out that most Markov models alone do not represent social network data well and have now generally been superseded by social circuit models. Nevertheless, a more detailed discussion of a few simple Markov models will assist understanding of the general modeling approach.

The simplest Markov model is the “2-star model,” where the higher-order star and triangle parameters are set to zero: $\theta_{S_3} = \theta_{S_4} = \dots = \theta_{S_{n-1}} = \theta_T = 0$. The configurations in the model are then simply edges and 2-stars.

The 2-star statistic may equivalently be written in terms of the degree distribution as $S_2(x) = \sum_{i < j} \sum_{k \neq i, j} x_{ik} x_{kj} = \sum_{j \geq 2} \binom{j}{2} d_j(x)$, where d_j is the number of nodes of degree j . For a given number of edges, a positive 2-star parameter increases the probability of graphs with more 2-stars. For a given number of edges, the simplest way to get many 2-stars is to have some high-degree nodes, thereby leading to a skewed-degree distribution. The degree variance $S = \frac{1}{n} \sum_{i=1}^n (x_{i+} - \bar{x})^2$ has been used as a measure of centralization (Hagberg, 2004; Snijders, 1981a, 1981b), and expanding this expression, the first term is a sum of 2-stars centered on node i , $x_{i+}^2 = \sum_{j,b} x_{ij} x_{ib}$. So, in fitting a 2-star model, we are modeling the average number of ties per node (the edge parameter) and the variation in the number of ties of each node (the 2-star parameter), that is, the mean and variance of the degree distribution. Some technical aspects of the 2-star model are studied more closely by Handcock (2003) and Park and Newman (2004).

Although this model accounts for degree variation, it rarely fits observed network data because it is unable to capture network closure (clustering or triangulation). However, the model can be extended by including triangle configurations. Frank and Strauss (1986) called this the “triad model.” Again, this model is of limited empirical value because it can only be fit to data under very limited circumstances. However, a brief discussion illustrates a number of important points. The model may equivalently be formulated in terms of triad counts (Frank & Strauss, 1986), that is, in counts of induced subgraphs consisting of three vertices with no, one, two, or three edges. Triad counts of undirected graphs have been extensively used to test hypothesis by comparing them against various null distributions (Frank, 1979; Frank & Harary, 1982; Holland & Leinhardt, 1971, 1976; Karlberg, 1997, 2002).

A positive triangle parameter indicates network closure or clustering. In friendship networks, for example, there may be a tendency for friends to meet through other friends (i.e., a friend of a friend is a friend). Because the 2-star configuration is a subgraph of the triangle, a lower-order interaction, it is desirable to include the 2-star parameter whenever the triangle parameter is included. Without a 2-star parameter, we cannot tell whether the prevalence of triangles was due to accumulation of 2-stars or a distinct effect of closure.

The triad model may be extended by including a 3-star parameter (Figure 6.4). Then, the edge, 2-star parameter, and 3-star parameter model the mean, standard deviation, and skew of the degree distribution,

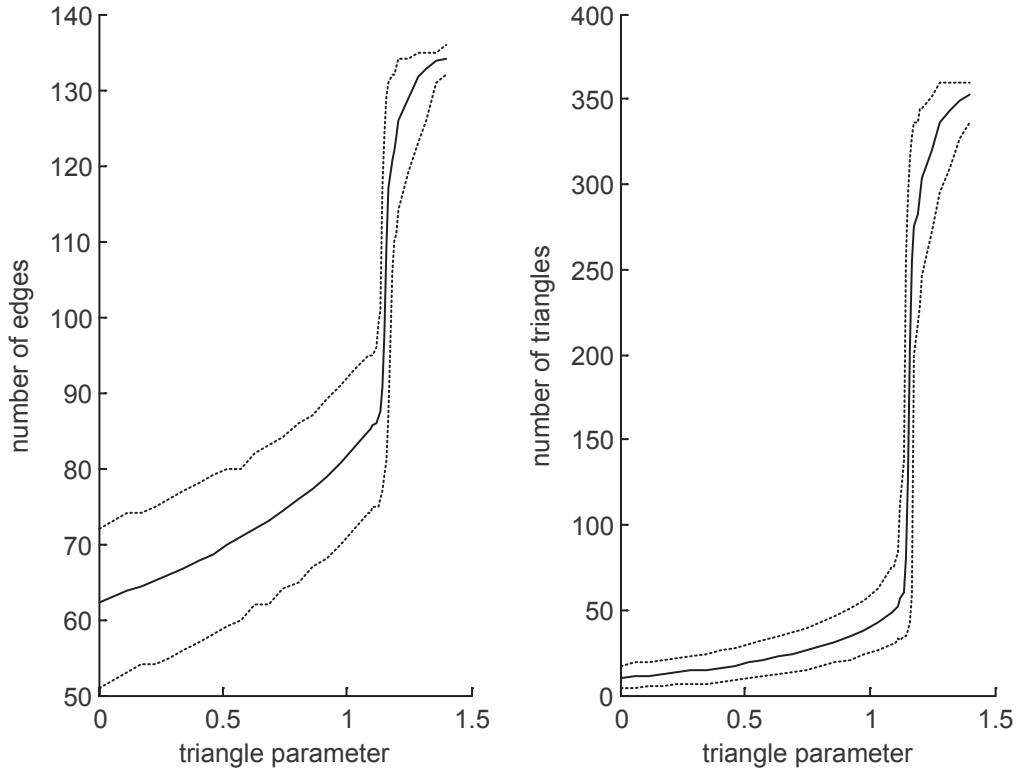


Figure 6.5. Expected and 95% intervals for number of edges and triangles as function of triangle parameter in Markov model on thirty nodes with edges (-3), 2-stars (0.5), 3-stars (-0.2), and triangles.

respectively. In terms of interpretation, positive 2-star and negative 3-star parameters indicate some centralization through high-degree nodes but with a cap on the level of that centralization (through the negative 3-star parameter). In other words, a positive 2-star parameter suggests a tendency to have multiple ties, but the negative 3-star parameter indicates that there is a point beyond which additional ties are not desirable (we cannot be friends with everyone!).

A fundamental property of an ERGM is that parameters regulate how many of their corresponding configurations are expected on average. As an illustrative example, consider how the triangle parameter θ_T controls the number of triangles in a model that also includes parameters for edges, 2-stars, and 3-stars, with parameter values fixed at -3.0 , 0.5 , and -0.2 respectively,³ for undirected graphs with thirty nodes. The expected number of triangles and the range of the likely number of triangles for different values of the triangle parameter is graphed in [Figure 6.5](#). If there is no effect for triangulation (i.e., $\theta_T = 0$) from the right-hand

³ For the purpose of simple illustration, we have chosen these numbers as parameter values based on experience.

panel, we expect around 10 triangles on average, and with probability .95, we observe between 4 and 17 triangles. As we increase the triangle parameter, the number of triangles we expect to see steadily rises: for example, when $\theta_T = 1$, we get 40 triangles on average, and 95% of the graphs generated from this model have between 25 and 59 triangles. Roughly after $\theta_T = 1.11$, there is a dramatic rise in the number of triangles, and when $\theta_T = 1.15$, the 95% interval extends from 38 to 293 triangles.

The statistics are highly interdependent: we cannot increase the expected number of triangles without also increasing the expected number of edges as in the left-hand panel. Hence, although we get between 51 and 72 edges when $\theta_T = 0$, we get between 71 and 92 edges when $\theta_T = 1$. The explosion in the number of triangles after $\theta_T = 1.11$ is mirrored in the steep rise in edges.

The transition from low-density graphs to high-density graphs in [Figure 6.5](#) illustrates one of the difficulties with Markov models. For models with a triangle parameter (all other parameters being fixed) in the range 0 to roughly 1.11, the increase in the number of triangles is small and gradual as a function of increases in the triangle parameter. In that range, the increase in the probability of a graph for an additional triangle is counteracted by the negative edge parameter (and also the negative 3-star parameter).

Compare two graphs, x and y , where y has one triangle more than x through the addition of an edge that closes a 2-path. Then y has an additional θ_T in the (exponent for the) probability compared to x , meaning that the triangle contributes positively to favoring y over x . However, an additional edge in y also contributes negatively to the probability of the graph given that the edge parameter is negative. Hence, leaving aside the star parameters for the moment, increasing the number of triangles contributes positively to the likelihood of observing that graph through the triangle parameter but contributes negatively through the edge parameter. In [Figure 6.5](#), we see that there is a balance between the positive triangle and negative edge effect until roughly $\theta_T = 1.11$. From roughly $\theta_T = 1.18$, the balance is upset in favor of graphs with many triangles. This dramatic change is known as a phase transition from low-density graphs to high-density graphs, characterized by high uncertainty with a very wide 95% interval.

[Figure 6.6](#) depicts the number of triangles across graphs when $\theta_T = 1.1524$. The high variability occurs because the model places half the probability on low-density graphs and half on high-density graphs, and very little on graphs in-between. Bimodality as in [Figure 6.6](#) is problematic for both inference and interpretation of parameters. Researchers have found that observed data often fall in between two such humps for Markov graphs, when it is impossible to find combinations of

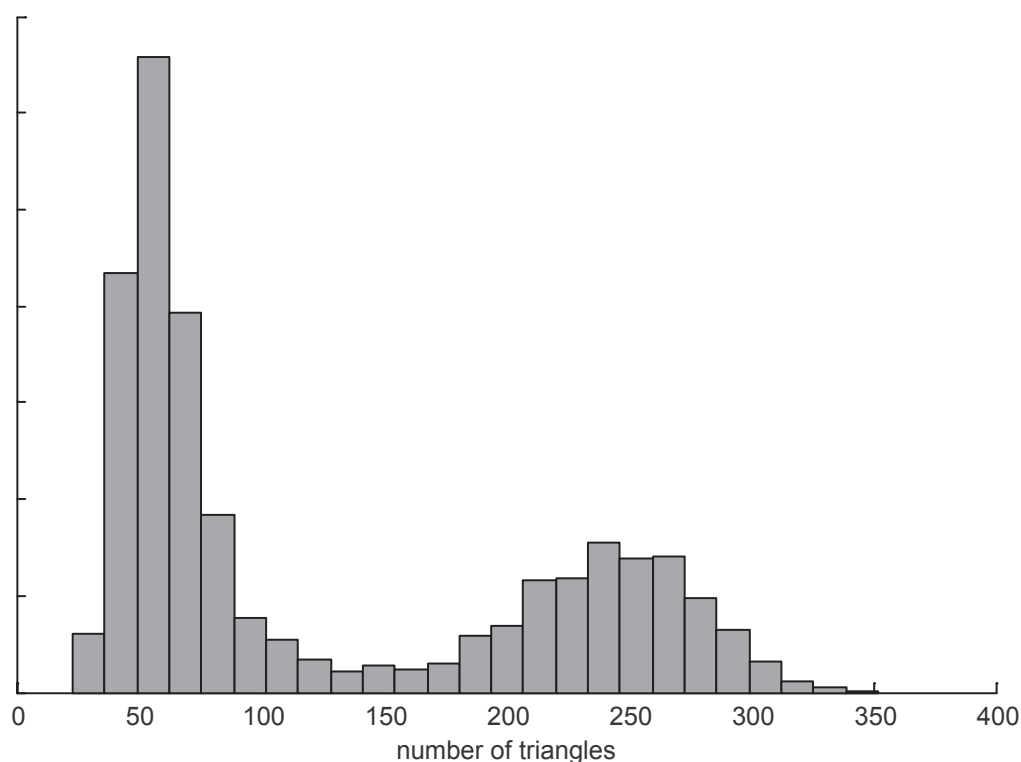


Figure 6.6. Number of triangles for Markov model with edges (-3) , 2-stars (0.5) , 3-stars (-0.2) , and triangles (1.1524) .

parameter values such that the observed combination of graph statistics is central in the distribution of graph statistics (see, e.g., Snijders et al., 2006).

Similar phase transitions can be observed for the star parameters, so although the triad model models the degree distribution and recognizes network closure, it has been demonstrated to be inadequate for reproducing the forms of degree heterogeneity and clustering commonly observed in empirical social networks (Handcock, 2003; Robins & Pattison, 2005; Snijders et al., 2006). It is perhaps not surprising given that social relationships are potentially complex phenomena that they cannot be described only at the dyadic and/or triadic levels implied by the Markov dependence assumption.

Alternating Star Parameters. One parameterization that helps considerably with phase transitions involving star parameters is that of “alternating star parameters” or (equivalently) “geometrically weighted degree parameters.” The alternating star statistic is a weighted sum of all star counts (with alternating signs on the counts – see the next paragraph) and dampens the effects of higher-order stars to help avoid a dramatic phase transition from low-density to high-density graphs.

When fitting models to real data, the combination of density, 2-star parameter, and 3-star parameter in the triad model often yields parameters with alternating signs (as do indeed models with higher-order stars). The alternating sign of star parameters is associated with the fact that the k -star counts are nested: every k -star contains $\binom{k}{j}$ j -stars, for $j \leq k$. Higher-order stars are higher-order statistical interactions in the model: changes in signs with higher-order interaction effects are familiar from standard general linear model approaches.

Frank and Strauss (1986) proposed to reparameterize the star parameters in terms of the degree distribution $\theta_{d(j)} = \sum_{k \geq j} \binom{j}{k} \theta_{S(k)}$, where $\theta_{d(j)}$ and $\theta_{S(k)}$ are parameters for nodes of degree j - and k -stars, respectively. This is a complete parameterization of the degree distribution but creates too many free parameters to be estimated. To limit the number of parameters, Snijders et al. (2006) suggested the restriction $\theta_{d_j} = \theta_{d_{j-1}} e^{-\alpha}$, for $\alpha > 0$, down-weighting the contributions of high-degree nodes in a geometrically decreasing fashion (see also Hunter (2007) for a slight variation). Here, α is a smoothing constant either fixed at a value chosen by the researcher or treated as a parameter to be estimated (see Chapter 12). These parameter restrictions for the degree parameters may be translated back to the parameters for the stars. With some minor modifications, $\theta_{d_j} = \theta_{d_{j-1}} e^{-\alpha}$ is equivalent to the constraint $\theta_{S(k)} = -\frac{\theta_{S(k-1)}}{\lambda}$ on the star parameters, where $\lambda = \frac{e^\alpha}{e^\alpha - 1}$. Treating λ as a fixed constant, this constraint implies that we have one alternating star parameter θ_S , with an associated statistic:

$$\begin{aligned} z_s(x; \lambda) = & S_2(x) - \frac{S_3(x)}{\lambda} + \cdots + (-1)^k \frac{S_k(x)}{\lambda^{k-2}} + \cdots \\ & + (-1)^{n-1} \frac{S_{n-1}(x)}{\lambda^{n-3}}. \end{aligned} \quad (6.5)$$

Because of the special properties of the stars of order 2 and 3, and their relation to degree variance and skew, it is sometimes useful to include effects for stars of order 2 and 3 in addition to the alternating star effects. It is almost always recommended to include the edge parameter θ_L in addition to the alternating star statistic (unless using fixed density models – see later in this chapter). The alternating star statistic differs from the geometrically weighted degree (GWD) statistic (Snijders et al., 2006) (and the subtly different GWD statistic of Hunter (2007)) in the treatment of 1-star (i.e., edges) (the exact relation is given in formula 14 of Snijders et al., 2006).

The general interpretation of the alternating star parameter θ_S is as follows. A positive parameter value with $\lambda > 1$ indicates centralization based on high-degree nodes. The value of λ controls the amount of that

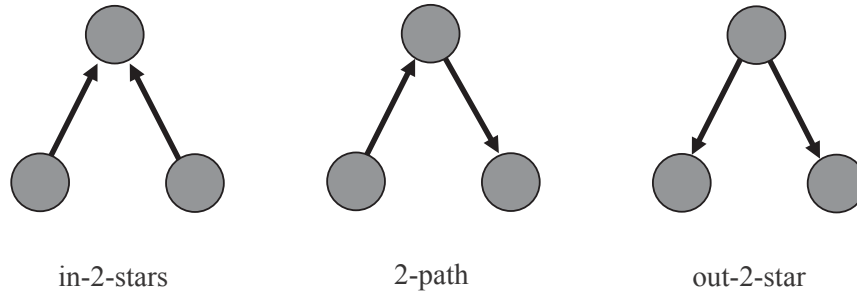


Figure 6.7. Directed star configurations on three nodes.

centralization, with a larger λ encouraging higher-degree nodes. Thus, a large λ may assist with modeling graphs with more highly skewed degree distributions. When θ_S is negative, however, the reverse will be true, so that the degree distribution is relatively more equal among nodes. For many empirical data sets, $\lambda = 2$ is a reasonable value for this smoothing constant for alternating statistics (Robins & Morris, 2007), although λ may also be considered a free parameter to be estimated (Hunter & Handcock, 2006; Snijders et al., 2006).⁴

Directed Markov Models. For directed graphs, there are many more configurations than for undirected graphs, too many to fit all of them in one model (see Figure 7.8 in Chapter 7 for the twenty-five configurations with less than four vertices). Some star configurations with special status in directed graph Markov models are in-2-stars, 2-paths, and out-2-stars, given by $IS_2(x) = \sum_i \binom{x_{+i}}{2}$, $P_2(x) = \sum_i \sum_{j,h,j \neq h} x_{ji}x_{ih}$, and $OS_2(x) = \sum_i \binom{x_{i+}}{2}$, respectively (shown in Figure 6.7). These are the configurations of three (non-isolate) nodes with two arcs (Snijders, 2002). Directed graphs have two degree distributions: the in-2-star parameter controls variation of in-degrees, whereas the out-2-star controls variation of out-degrees. So, in principle, the in-2-star and out-2-star parameters model the heterogeneity in a graph with respect to popularity and activity. The 2-path parameter controls for the correlation between in- and out-degree.

The full Markov model for directed graphs parameterizes the entire triad census (Holland & Leinhardt, 1970), although again this is generally too many parameters. If we limit the triangle parameters to those that consist of three vertices with exactly one arc between any pair of nodes,

⁴ As Snijders et al. (2006) point out, if λ is considered a free parameter, the model no longer belongs to the exponential family of distributions but to the curved exponential family of distributions. Hunter and Handcock (2006) show how λ may be estimated using maximum likelihood, and Koskinen, Robins, and Pattison (2010) estimate the smoothing constant using Bayesian inference (see Chapter 12). This also holds for smoothing constants in the other alternating statistics presented here.

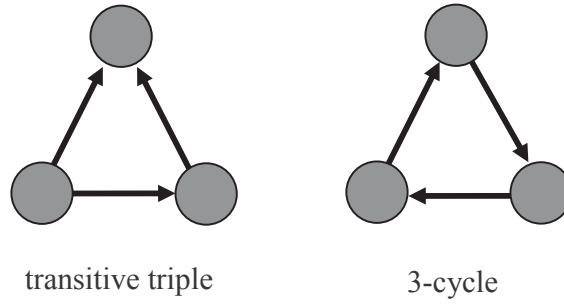


Figure 6.8. Configurations on three vertices with exactly one tie for each dyad.

the configurations correspond to the 3-cycle and the transitive triple (Figure 6.8). The 3-cycle can be seen to consist of three distinct 2-paths, and the transitive triple consists of a 2-path, in-2-star, and out-2-star. These are effects that should be controlled for when 3-cycle and transitive triadic parameters are included in a model. In addition, a directed Markov model should include the arc and reciprocity parameters familiar from the dyadic dependence directed model, as well as star-based parameters to control for the degree distribution.

Transitivity and cyclicity represent different forms of network closure. They have interesting differences in regard to local hierarchies in the network. Consider the case when the network represents the flow of orders, so that a node may only send ties to other nodes of lower status. In that case, there can be no cycles, but there may be transitive triples. (Local hierarchy, however, does not necessarily imply global hierarchy.) It is quite common in friendship networks to see a negative effect for 3-cycles and a positive effect for transitive triads, suggesting that friendship is quite strongly organized in terms of local hierarchies. Sometimes cyclicity is also interpreted as a form of generalized exchange.

Even though the family of Markov models covers a variety of different models and is able to produce many different types of graphs (Robins, Pattison, & Woolcock, 2005), difficulty in fitting models to real data as described previously (Corander et al., 1998; 2002; Frank & Strauss, 1986; Handcock, 2003; Snijders, 2002) prompted the introduction of statistics that were not derived exclusively from Markov dependence.

We have already introduced alternating star statistics for undirected models, motivated by these issues. Alternating in- and out-star parameters can be used in directed models in an analogous way. Two alternating counts of in- and out-degrees suggested by Snijders et al. (2006) were

$$z_{d,out}(x; \alpha) = \sum_{j=0}^{n-1} e^{-\alpha j} d_j^{out}(x) = \sum_{i=0}^n e^{-\alpha x_i} \quad (6.6)$$

and

$$z_{d,in}(x; \alpha) = \sum_{j=0}^{n-1} e^{-\alpha^j} d_j^{in}(x) = \sum_{i=0}^n e^{-\alpha x_{+i}}. \quad (6.7)$$

These are referred to as “geometrically weighted out-degrees” and “geometrically weighted in-degrees,” respectively. In the same fashion as we may express the geometrically decreasing degrees statistic as alternating star statistics, geometrically weighted out- and in-degrees have an alternate representation as alternating out- and in-stars, respectively (Robins, Pattison, & Wang, 2009). As $z_{d,out}(x; \alpha)$ models the shape of the out-degree distribution, the corresponding effect may be called “activity spread.” Analogously, we term the effect corresponding to $z_{d,in}(x; \alpha)$ “popularity spread,” referring to the way in which differences in number of ties received (for positive relations) reflect differences in popularity. A large positive value on the in- or out-star parameter indicates a network with high- (in- and out-) degree nodes (i.e., a network more centralized in terms of popularity or activity, respectively). The parameter α is usually treated as a smoothing constant for alternating statistics (as described previously).

The social circuit dependence assumption permits further extensions to address the problems associated with fitting Markov models.

6.6.4 Social Circuit Models

Undirected Models. The social circuit dependence assumption, in combination with Markov dependence, leads to the following configurations: edges, k -stars of different order, k -triangles, and k -independent 2-paths. These are the most commonly used social circuit configurations in model fitting (although the social circuit assumption also permits other possible configurations, such as cliques of different sizes, that are not often used).

A k -triangle consists of two connected nodes that are also jointly connected to k other distinct nodes. A k -independent 2-path consists of two nodes jointly connected to k other distinct nodes. These are depicted in [Figure 6.9a](#) (k -triangles) and [Figure 6.9b](#) (k -independent 2-paths), where we refer to (i, j) as the *base* and (i, h_k) and (j, h_k) as the *sides* of the k -triangle. Note that these configurations are consistent with the combined social circuit/Markov dependence assumptions because they are counts of 4-cycles (social circuit) with – for k -triangles – a base that is accounted for by Markov dependence.

The motivation for using k -triangle configurations relates to the difficulty in modeling triangles in Markov models as noted previously. A Markov model, in effect, assumes that triangles are distributed evenly

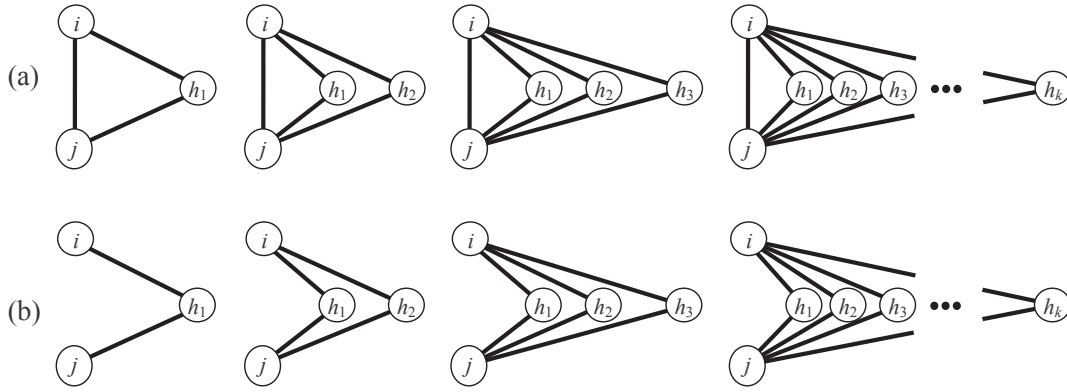


Figure 6.9. (a) Alternating triangles on base (i, j) and (b) independent 2-paths.

throughout the graph. For many observed networks, however, triangles are clumped together in denser regions of multiple triangulation, which the higher-order k -triangles are better able to model.

Just as a 2-star and 3-star model the variance and skew of the degree distribution, the 2-triangle and 3-triangle model the variance and skew of the distribution of triangles across the graph – more specifically, the distribution across ties (i, j) of the number of third nodes connected to both i and j (termed the “edgewise shared partner distribution”).

The k -2-path configuration is lower order to the counterpart k -triangle configuration, so the inclusion of both parameters permits inference about whether k shared partners encourage the presence of a tie between two nodes. (The distribution of k -2-paths across dyads is often referred to as the “dyadwise shared partner distribution.”)

The count of k -triangles in a graph x is $T_k(x) = \sum_{i < j} x_{ij} \binom{S_{2ij}(x)}{k}$, where $S_{2ij}(x) = \sum_{k \neq i, j} x_{ik} x_{kj}$ is the number of 2-paths connecting i and j (except for the 1-triangle, which is counted as the number of triangles, as in the Markov case). As with the alternating star statistic, distinct parameters for each of the k -triangles and k -2-paths give an overdetermined model with too many parameters. Snijders et al. (2006) proposed to combine the counts of k -triangles through the alternating triangle statistic, analogously to the alternating star statistic:

$$z_T(x; \lambda) = 3T_1(x) - \frac{T_2(x)}{\lambda} + \frac{T_3(x)}{\lambda^2} + \dots + (-1)^{k-1} \frac{T_k(x)}{\lambda^{k-1}} + \dots + (-1)^{n-3} \frac{T_{n-2}(x)}{\lambda^{n-3}}. \quad (6.8)$$

(The factor of 3 for the T_1 count arises because the 1-triangles are counted as single Markov triangles – see Snijders et al. (2006).) The constraints

on the k -triangle parameters implied by the alternating triangle statistic means that edges whose nodes are connected by many indirect paths get differentially weighted. The strength of this effect will depend on the values of the parameter and the damping factor λ . For $\lambda > 1$ and a positive parameter value, the model favors graphs with highly clustered regions. The weight of higher-order k -triangles is smaller the greater the value of λ . For higher values of λ , the highly clustered regions of the graph tend to be larger. So, the alternating triangle statistic represents network closure in dense regions of the network with the size of those regions represented by the value of λ . (In practice, we typically set $\lambda = 2$ and treat it as a smoothing parameter for alternating statistics, as explained previously.) The geometrically weighted edgewise shared partner (GWESP) term in *statnet* is parameterized in terms of $\log(\lambda)$, so that `gwesp(0.693)` is the alternating triangle statistic with $\lambda = 2$. The geometrically weighted dyadwise shared partner (GWDSP) statistic (alternating 2-paths) is similarly parameterized, so that `gwdsp(0.693)` has $\lambda = 2$.

In the Markov model with a positive triangle parameter, the triangles are “evenly” distributed in the graph. In a model with an alternating triangle statistic, however, the presence of one triangle may increase the likelihood of other triangles “attaching themselves” to it. However, the increase in probability for creation of a k -triangle is a decreasing function of k . This can be interpreted as follows: if a social tie between individuals who share many social partners is absent, the increase in a number of shared partners is not likely to increase the probability of the tie greatly. In empirical settings, we may see serious impediments to a tie being formed (e.g., mutual antipathy or geographic distance), and additional shared partners will not alter the situation much.

It is obvious that alternating triangles emerge not only from formation of new bases of k -triangles but also from the formation of new sides. Analogous to the use of 2-paths as a control for triangles that result from accumulated two paths, the alternating independent 2-path statistic can be used as a control for the alternating triangle statistic. As previously, a sequence of alternating and decreasing weights for the counts of independent 2-paths (Figure 6.9b) can be used to create a single aggregate statistic termed the “alternating path statistic” with a damping factor λ .

Note that an independent 2-path of order 2 corresponds to a 4-cycle. The constant λ may be distinct from that for alternating triangles (although we do not make any notational distinctions), but again it is commonly set to 2. There is a functional dependence between alternating 2-paths and alternating triangles, and it is common that the 2-path parameter is negative in combination with a positive triangle parameter. In that case, actors are less likely to have shared partners unless they are themselves connected, a clear indicator for network closure.

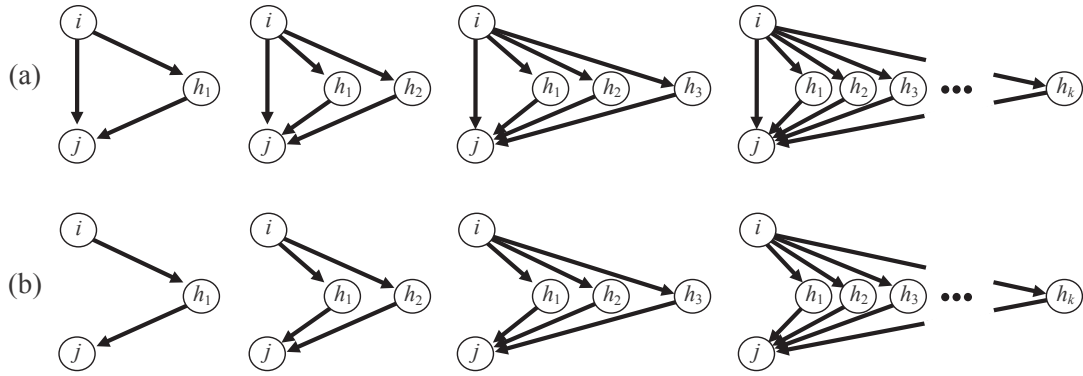


Figure 6.10. Configurations for directed graphs in alternating forms (a) AT-T and (b) A2P-T.

Directed Graphs. Just as with Markov models, the directed equivalent of the social circuit specifications has more variety in possible configurations. We have already mentioned parameters for alternating in- and out-stars.

For a directed version of the alternating triangle configuration, Snijders et al. (2006) proposed the transitive version shown in Figure 6.10a. The vertices at the base, i and j , are connected to each other and also indirectly through 2-paths via k others. The lower-order k -2-path counterparts are depicted in Figure 6.10b. As in the undirected case, the number of parameters can be reduced by using alternating parameter constraints. These alternating statistics are denoted AT-T and A2P-T, respectively, where T signifies “transitive.” (For directed networks, the GWESP and GWDSP terms in *statnet*, `gwesp()`, and `gwdsp()` are equivalent to these transitive configurations AT-T and A2P-T with the $\log(\lambda)$ parameterization.) The counterpart parameters are referred to as the “path closure” and “multiple 2-path” parameters, respectively.

Considering the way the directed k -triangles are built on the directed 1-triangle (i.e., the transitive triple), we note that there are other possible choices of base. In an undirected triangle, the three vertices are structurally equivalent, so that the subgraph stays the same even if the labels of the nodes are rearranged. In a transitive triple, however, the three nodes have distinct positions: one sends two ties, one receives two ties, and one receives and sends one tie. In the directed 1-triangle in Figure 6.10a, these roles are occupied by i , j , and h_1 , the center of an out-2-star, an in-2-star and a 2-path, respectively. Different directed k -triangles may be formed by adding out-2-stars, in-2-stars, and 2-paths to a suitable base. In addition to the AT-T structure in Figure 6.10 (formed by adding 2-paths to a base), Robins et al. (2009) proposed the directed k -triangles in Figure 6.11: (1) AT-U, by adding 2-out-stars to the base; (2) AT-D, by adding 2-in-stars; and (3) AT-C to form cyclic patterns. The 2-path “controls” for the latter two are given by the alternating 2-paths in Figure 6.12: A2P-U (“shared activity”) and A2P-D (“shared popularity”).

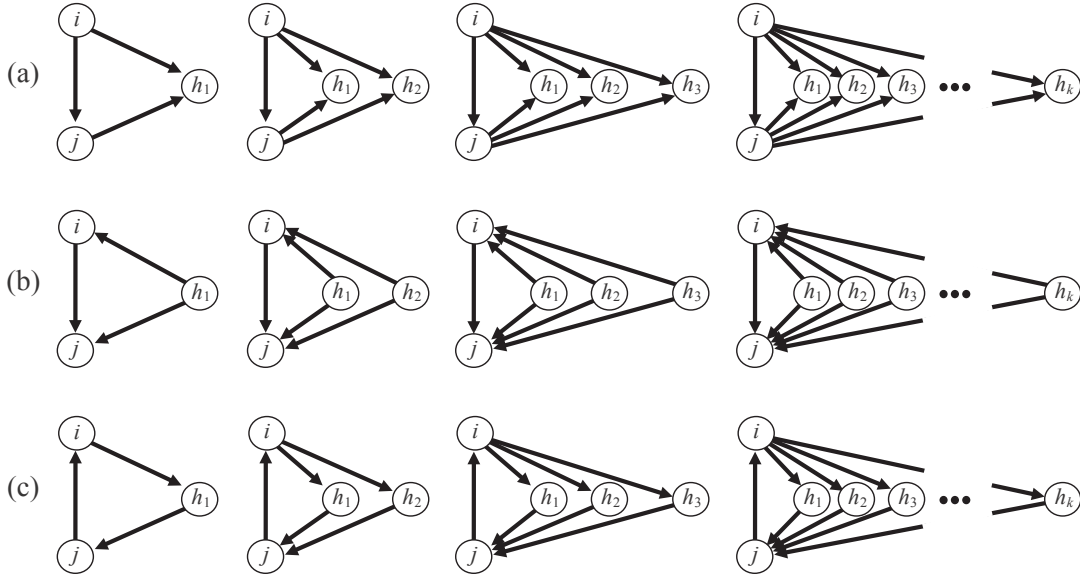


Figure 6.11. Additional triadic configurations for directed graphs in alternating forms (a) AT-U, (b) AT-D, and (c) AT-C.

These four different forms of alternating triangle effects relate to different forms of network closure. (The notation in Figure 6.11 of AT-D and AT-U derives from the classic triad census, with the D and U referring to “down” and “up”; the AT-C notation refers to “cycle.” See Chapter 15 for an example of the application of such configurations.)

We may think of AT-T in terms of transitive path closure. In terms of friendship, i tends to choose those j who are friends of their friends h_u . Following Robins, Pattison, and Wang (2009), a positive AT-T parameter, the “path closure” parameter, can be interpreted as a tendency for structural holes to close when there are multiple independent paths between i and j .

A positive AT-U parameter – referred to as an “activity closure” parameter – suggests that if two actors i and j are similar with respect to

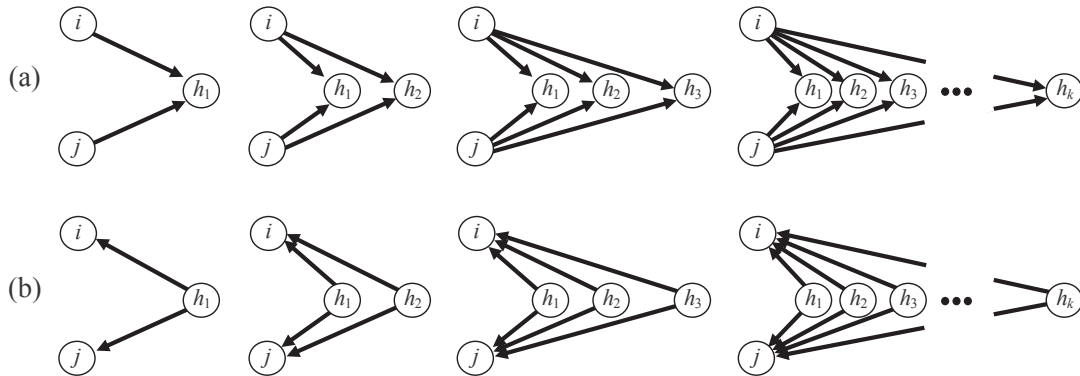


Figure 6.12. Additional 2-path configurations for directed graphs in alternating forms (a) A2P-U and (b) A2P-D.

activity – that is, they send ties to the same people – then this increases the probability that there is a tie between them. Implicit in this interpretation is that if individuals send ties to the same others, then they are either doing this because they are faced with the same conditions or embedded in the same social neighborhoods or social settings. These conditions may, of course, be created or reinforced by repeated similarity of action. For example, we could think of two individuals new to a group who therefore send ties to popular people and, in doing so, recognize that they are both in a similar position, something that is acknowledged by the establishment of a tie between them.

A positive AT-D parameter – a “popularity closure” parameter – may be interpreted in similar terms, but in this case for actors who are similar with respect to popularity rather than activity. One interesting example would be if the relation investigated is positive, yet there was a degree of competition among highly connected nodes, in which case this competitiveness might act against the prevalence of AT-D counts. A positive AT-C parameter – a “cyclic closure” parameter – indicates the presence of many 3-cycles in the network. This represents nonhierarchical network closure.

Robins, Pattison and Wang (2009) also proposed that equating the three transitive triadic parameters – path, activity, and popularity closure – to create one *general transitivity* parameter might also be useful and parsimonious in certain circumstances (we label the associated statistic “AT-TDU”). A positive general transitivity parameter indicates a general tendency for hierarchical network closure without differentiating among the three effects. Of course, it is an empirical question whether a particular network exhibits this general effect or whether one or other of the individual transitive closure effects is paramount.

Robins, Pattison and Wang (2009) also suggested inclusion of three parameters that can help control the degree distributions of directed graphs. These were parameters for the number of “isolated nodes” (i.e., nodes with degree zero), for “sources” (i.e., nodes with zero in-degree but positive out-degree), and for “sinks” (i.e., nodes with zero out-degree but positive in-degree).

Interpreting parameters and their effects independently in isolation from the other effects may be hazardous in light of the high degree of interdependence between different counts. The magnitudes of the estimated parameters are typically hard to interpret as well. Accordingly, although we may provide interpretations along the lines suggested in previous paragraphs, it is often helpful to interpret the model – that is, the simultaneous results of the parameters in the model – in terms of the kinds of structure produced through simulation (see [Chapter 12](#)). Sometimes it is helpful to use conditional log-odds and odd ratios, especially when we include actor attributes in models, as we illustrate in [Chapter 8](#).

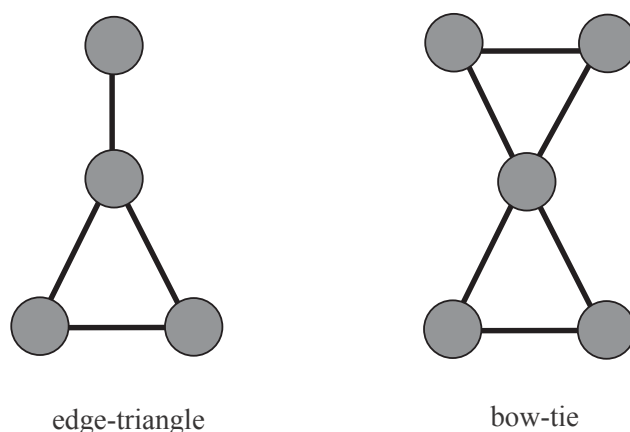


Figure 6.13. Configurations associated with brokerage.

6.7 Other Model Specifications

Whether a particular model specification is appropriate in a given context is in part an empirical question, regardless of how coherent the underpinning theoretical framework. Almost certainly, therefore, we will come to entertain different dependence assumptions and different models within the broad ERGM class as we work with new forms of network data, and as we gain experience in understanding the strengths and limitations of particular model specifications.

Butts (2006) proposed a form of dependence termed “reciprocal path dependence,” the idea that two tie-variables are conditionally dependent if their presence creates a directed cycle of any length in the graph. Statistics for the resulting homogeneous model are the counts of cycles of each length in the graph, the so-called cycle census.

Although closure is testified as being a central mechanism in tie formation, we saw that too much closure is responsible for the instabilities of the pure Markov model (Figures 6.5 and 6.6). Robins, Pattison, Snijders, and Wang (2009) also argued that as there are costs associated with establishing and maintaining relationships with people, there may be additional costs associated with maintaining big groups. Consequently, both from the perspective of creating models that generate realistic graphs and based on theoretical considerations, tendencies for nonclosure might be considered. Robins, Pattison, Snijders, and Wang (2009) proposed configurations such as edge-triangles and bow ties, representing particular mechanisms that counteracted closure (Figure 6.13). The edge-triangle configuration is interpreted as a form of brokerage from the central node (the one with three ties). Similarly, the bow tie may be seen to represent partially overlapping group membership, where there is one node that belongs to two triangles. Here, the triangle is interpreted as the simplest

form of a group. There is ongoing work examining the complexities of these additional configurations.

In these cases, a means of building model specifications from a particular construal about dependencies has provided a valuable path to model building. Equally important, though, in an empirical setting, is the means of evaluating how adequately a model has been specified. This is an issue that is taken up in [Chapter 12](#).

6.8 Conclusion

The central premise of the ERGM modeling endeavor is its dual interpretation as a model for ties and for the graph. This duality mirrors both theoretical considerations – the way individuals form ties but at the same time are constrained and affected by structure – and technical considerations – we provide a model for tie-variables but conditional on the rest of the graph. In this chapter, we define different model specifications in terms of localized structures and attempt to interpret these in terms of substantively interesting effects. In [Chapter 7](#), we show how the dual consideration for ties and overall structure give rise to and motivate the inclusion of the subgraph counts presented in this chapter. We explain how the dependence assumptions relate to subgraphs and how the probability of a tie is informed by neighboring structures.

Dependence Graphs and Sufficient Statistics

Johan Koskinen and Galina Daraganova

7.1 Chapter Outline

This chapter is written for those interested in a more detailed understanding of how assumptions regarding various forms of dependence can be formalized. What is treated here is not essential for applying exponential random graph models (ERGMs) and may be skipped at a first reading. The general idea, laid down by Frank and Strauss (1986), is nevertheless crucial to the formulation of statistical models treated in this book.

Important key points in this chapter are as follows:

- Subgraph counts are not arbitrarily chosen in ERGMs but correspond to specific dependency structures.
- The subgraph counts in ERGMs are intricately nested and interdependent, so care has to be taken in interpreting parameters in isolation.
- An ERGM is akin to a log-linear model where the subgraph counts are represented by interactions of tie-variables.
- ERGMs try to reduce the complexity of observed networks into systematic underlying principles and stochastic components.
- A homogeneous ERGM assigns equal probability to graphs that are structurally identical.

In this chapter, we focus on models for undirected graphs. Dependence graphs for directed models are a natural extension of what we describe here, but we only discuss them briefly.

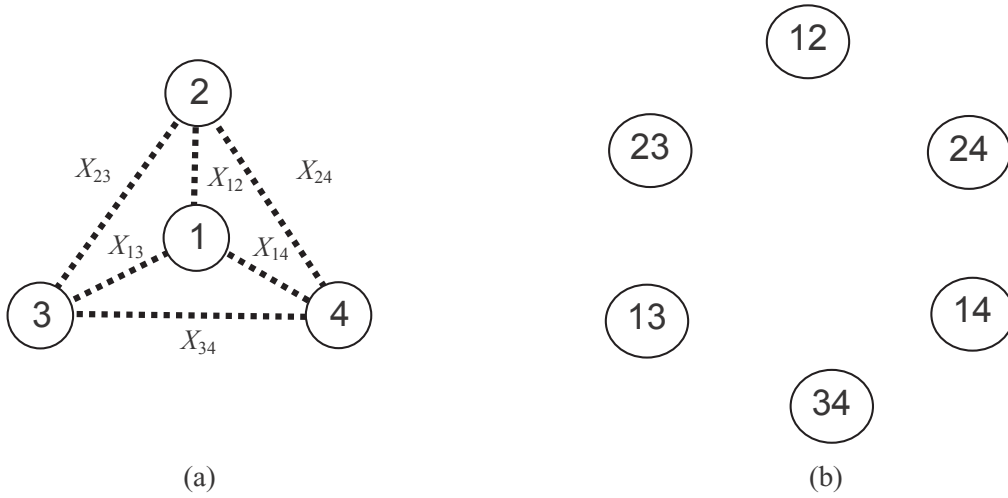


Figure 7.1. Tie-variables of (a) four-node graph and (b) associated Bernoulli dependence graph.

7.2 Dependence Graph

Frank and Strauss (1986) proposed a dependence graph D to represent the dependence assumptions for tie-variables.¹ The dependence graph is a representation of the way in which tie-variables may depend on one another.

Formally, a dependence graph D for a (random) graph G is defined using the tie-variables X_{ij} from G . D has as its node set the tie-variables X_{ij} so that there are $n(n-1)/2$ and $n(n-1)$ nodes in D according to whether G is undirected or directed. The edges of D , however, represent the conditional dependencies among the tie-variables of G . That is, (X_{ij}, X_{kl}) is an edge of D if and only if X_{ij} and X_{kl} are assumed to be conditionally dependent given the values of all remaining tie-variables. Hence, the dependence graph indicates which random variables are conditionally dependent on (or conditionally independent of) each other. It is worth emphasizing that the dependencies among random variables are indeed a matter of assumption. They reflect the basic hypotheses that the researcher assumes underpin the model.

Take as a simple example possible networks on only four nodes. With four nodes, there are exactly six tie-variables (Figure 7.1a). Thus, the dependence graph will have six nodes, one for each tie-variable. For instance, the tie-variable X_{23} in Figure 7.1a is represented by node 23 in the dependence graph.

¹ Frank and Strauss derived their dependence graph from similar notions in the graphical modeling literature (see Lauritzen, 1996). In that literature, the same construct is termed an “independence graph.”

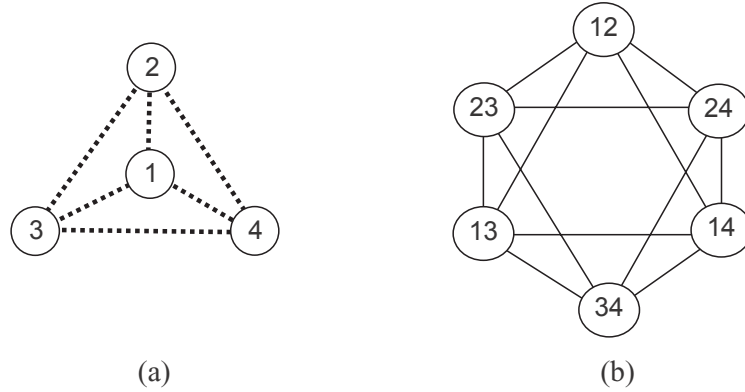


Figure 7.2. Tie-variables of (a) four-node graph and (b) associated Markov dependence graph.

Suppose we want D to represent the Bernoulli dependence assumption, where tie-variables are independent of each other. An edge in the dependence graph indicates a conditional dependence between two tie-variables, so in this case there are no edges in D . The dependence graph for the Bernoulli assumption is an empty graph as in Figure 7.1b.

Suppose, instead, that the dependence graph D represents Markov graph dependence. Then two tie-variables X_{ij} and X_{hm} are dependent, given the rest of X , if the intersection of $\{i,j\}$ and $\{h,m\}$ is nonempty. Edges in D are between tie-variables that share a node as in Figure 7.2b (e.g., there is an edge between nodes 12 and 24 because they have node 2 in common).

A very important feature of this dependence graph is that although there are edges between many nodes in Figure 7.2b, D is not complete (e.g., there is no edge between nodes 12 and 34). What this means is that, for a given graph G , once we know the rest of the graph (i.e., the values of the tie-variables X_{ij} for every pair of nodes excluding 12 and 34), then knowing X_{12} does not help us determine X_{34} : the two variables are conditionally independent (i.e., conditional on the rest of the graph). There must be some level of independence or else the models will not be identifiable and cannot be estimated from data.

For the social circuit model, the dependence assumption states that, in addition to Markov dependencies, two tie-variables X_{ij} and X_{hm} are conditionally dependent even when the intersection of $\{i,j\}$ and $\{h,m\}$ is empty, as long as ties between i and h and between j and m exist in the observed graph G . This assumption may be described in terms of the 4-cycle as explained in previous chapters. This complicates the representation of a dependence graph because some edges may appear due to the values of tie-variables in G . Note, however, that the social circuit models also presuppose Markov dependence, so the dependencies in Figure 7.2b at the least are present. In Figure 7.2b, there is no edge

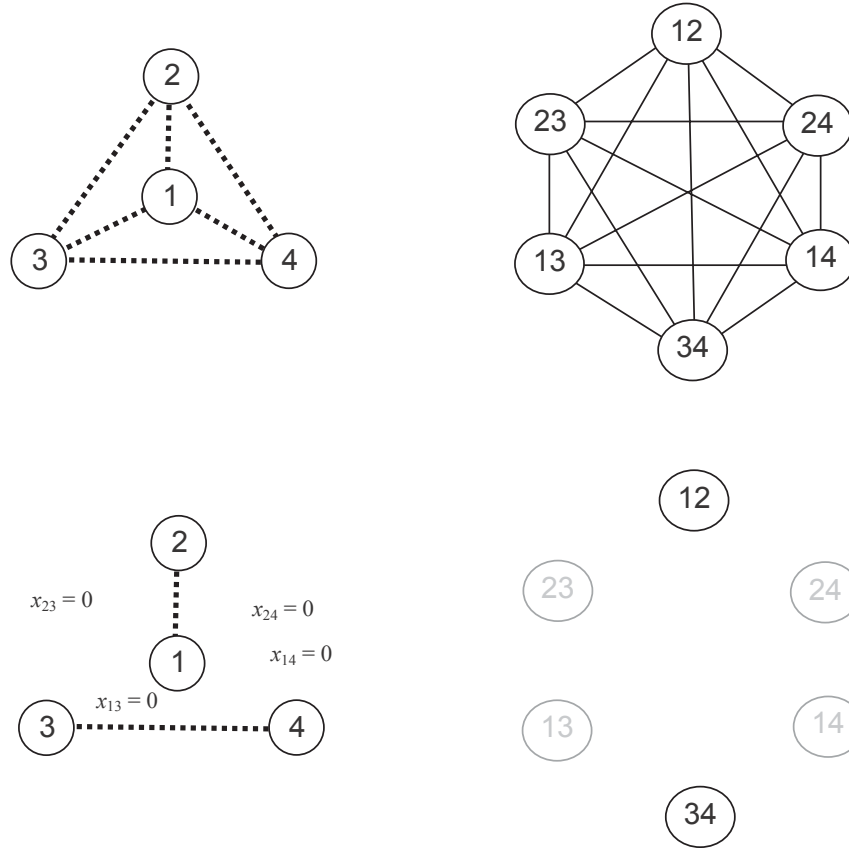


Figure 7.3. Tie-variables of social circuit graph and its dependence graph (top) and dependence graph conditional on some tie-variables being zero (greyed out vertices not in partial dependence graph).

between 12 and 34, but in social circuit dependence there may be an edge between 12 and 34 if, for example, $x_{13} = x_{24} = 1$ because then a 4-cycle can be created. If, however, all tie-variables other than X_{13} and X_{24} are zero, we know that the ties $\{1,2\}$ and $\{3,4\}$ could not create a 4-cycle, and in that scenario X_{13} and X_{24} are independent. So, there is some level of independence available for an identifiable model; this may be specified in the “partial dependence graph” conditional on some variables being zero, $x_{13} = x_{14} = x_{23} = x_{24} = 0$ (in Figure 7.3). In this dependence graph, if we know that $x_{13} = x_{14} = x_{23} = x_{24} = 0$ – represented in Figure 7.3 by the lighter shading of the nodes 13, 14, 23, and 24 – then X_{13} and X_{24} are independent, and hence there is no edge between 12 and 34. In principle, there is a partial dependence graph describing the dependencies that exist when each possible subset of tie-variables is set to zero, so that the complex social circuit dependence structure (for any G) can – again, in principle – be represented by a series of partial dependence graphs. Once again, we see that, as for the Markov dependence graph, there are pairs of tie-variables in the partial dependence graph that are

not tied. Consequently, there are instances of conditional independence. Partial conditional dependence was introduced for a number of different conditions for ERGMs in Pattison and Robins (2002), where further technical details can be found.

We have now constructed dependence graphs for three dependence assumptions. Frank and Strauss (1986) showed that the dependence graph is a conceptual link between the dependence assumptions and how they are expressed in the data. Although the dependence assumption implies that a network can be characterized by a limited number of subgraphs, the corresponding dependence graph delineates the subgraphs of particular importance.

In summary, the link between the dependence graph and the model is that collections of nodes in D that are mutually connected to each other – that is, cliques of tie-variables in D – can be represented in the model by the corresponding interaction effects among tie-variables. Each clique in the dependence graph represents a possible configuration and hence a possible parameter in the model. More important, if there is no clique in D for a set of nodes, then there should not be a configuration involving that set of nodes in the model.

For example, if the two tie-variables X_{12} and X_{13} are linked in D , then a model for the network that respects the dependence assumption underlying D may include the interaction effect $X_{12}X_{13}$. If X_{12} and X_{13} are linked in D , then X_{12} and X_{13} are conditionally dependent given the rest of G . By the definition of dependence, $\Pr(X_{12} = x_{12}, X_{13} = x_{13} | \text{rest}) \neq \Pr(X_{12} = x_{12} | \text{rest}) \Pr(X_{13} = x_{13} | \text{rest})$, so that the ties 12 and 13 would co-occur with a probability different from expected under the marginal probabilities for 12 and 13. The implication is that there may be an interaction effect $X_{12}X_{13}$. Furthermore, suppose that X_{12} , X_{13} , and X_{23} were all jointly dependent given the rest of the graph (as they would be under a Markov assumption), then the extent of the co-occurrence of these three tie-variables would be measured by the interaction $X_{12}X_{13}X_{23}$.

Note that if $x_{12} = x_{13} = 1$, then $x_{12}x_{13} = 1$, and there is a 2-star on node 1. Thus, the interaction effect $x_{12}x_{13}$ represents a possible 2-star, a Markov configuration, and so a parameter in the Markov model. Furthermore, if $x_{12} = x_{13} = x_{23} = 1$, then $x_{12}x_{13}x_{23} = 1$, and there is a triangle on nodes 1, 2, and 3. Hence, the interaction effect $X_{12}X_{13}X_{23}$ represents a possible triangle, another Markov configuration, and so another parameter in the Markov model. It is in this way that the cliques of the dependence graph come to identify possible effects in the model. Suppose, however, that $x_{12} = x_{34} = 1$, then although it is true that $x_{12}x_{34} = 1$, there is no clique for 12 and 34 in D because there is no edge between them. So, *for a Markov model*, it would be improper to include a parameter for this configuration of ties.

7.2.1 Hammersley-Clifford Theorem and Sufficient Statistics

The important link between the probability of a graph and the dependence graph is provided by the Hammersley-Clifford theorem (Besag, 1974). The theorem is an important basis for the graphical modeling literature (e.g., Lauritzen, 1996) and was introduced into network statistical methodology by Frank and Strauss (1986). The main points are the following (we have left out the details, which can be found in Frank and Strauss (1986) and Robins and Pattison (2005)).

The probability distribution function for any graph distribution may be written $\Pr(X = x) = \kappa^{-1} \exp Q(x)$ for some real-valued function Q (where κ is a normalizing constant and assuming $\Pr(X = x) > 0$ for all $x \in \mathbf{X}$). By writing $Q(x) = \log\{\Pr(X = x) / \Pr(X = x^{(0)})\}$ with reference to the empty graph $x^{(0)}$, we can use the inclusion–exclusion principle to write $Q(x) = \sum \lambda_A(x_A)$ for functions λ_A defined on subsets $A \subseteq J$. For a ratio $\Pr(X = \Delta_{ij}^+ x) / \Pr(X = \Delta_{ij}^- x)$, it follows from the definition of conditional probability that $Q(\Delta_{ij}^+ x) - Q(\Delta_{ij}^- x)$ may be a function only of x_{kl} that x_{ij} is *conditionally dependent on* – these x_{kl} are given by D . Hence, it can be seen (although this may require some thought) that $\lambda_A(x_A) \neq 0$ only for A that are cliques of D .

To arrive at the sufficient statistics, note first that for a binary graph we may write $\lambda_A(x_A) = \theta_A \prod_{\{i,j\} \in A} x_{ij}$. Because $\prod_{\{i,j\} \in A} x_{ij}$ is a product of binary variables, it can only take two values: 1 if all the variables in the index set A are equal to one, and 0 if at least one of the variables in the index set A is equal to zero. We may call A a configuration because it corresponds to a subset of potential ties, and $\lambda_A(x_A) = \theta_A$ when all ties of the configuration A are present. If, for example, A is equal to $\{i,j\}, \{j,h\}, \{h,i\}$, A is present if the triangle involving nodes i , j , and h is present (i.e., if $x_{ij}x_{jh}x_{hi} = 1$ or x_{ij} , x_{jh} , and x_{hi} are all equal to one). The function λ_A is called the A interaction in X .

The model thus expressed in terms of $\lambda_A(x_A) = \theta_A \prod_{\{i,j\} \in A} x_{ij}$ has the form of an exponential random graph model as presented in Chapter 6 (Equation 6.2). In other words, the Hammersley-Clifford theorem completely determines the general form of the model once a dependence assumption is adopted. In these network terms, the theorem can be expressed as

$$\begin{aligned} \Pr(X = x) &= \kappa^{-1} \exp \sum_{A \subseteq J^*} \theta_A \prod_{\{i,j\} \in A} x_{ij} \\ &= \kappa^{-1} \exp \sum_{A \subseteq J^*} \theta_A z_A(x), \end{aligned} \tag{7.1}$$

where J^* is the set of all cliques of the dependence graph, θ_A is a parameter associated with the clique (configuration) A , and $z_A(x)$ is an indicator

variable that has value 1 if configuration A is observed in graph x , and is 0 otherwise.

It is worth noting that any single tie-variable is also a clique in the dependence graph, and any subgraph of a clique is also a clique (except for certain partial dependence structures). Hence, if A is a clique of D , then so is every subset of A ; consequently, if A is a possible configuration in the model, so are subgraphs of A .

7.2.2 Sufficient Subgraphs for Nondirected Graphs

Bernoulli Model. For the Bernoulli model, the dependence graph (Figure 7.1b) is an empty graph, and hence the only cliques are those comprising a single node from D , each of which relates to an edge in the graph x . So, a model given by Equation (7.1) has a unique parameter for each tie-variable, a large number of parameters. For example, for a network with 20 nodes, the model has $\binom{20}{2} = 190$ parameters to be estimated.

One way of reducing the number of distinct configurations, A , and corresponding parameters, θ_A , is to assume that all structurally identical graphs are equally probable (i.e., that only the structure is relevant). This homogeneity assumption is a central ingredient in modeling networks using ERGMs.

For example, let the parameter θ_{ij} correspond to the edge configuration $A = \{i, j\}$. For a homogeneous model, we equate the parameters $\theta_{ij} = \theta_L$ for all i, j . From Equation (7.1), we then have that

$$\begin{aligned} \Pr(X = x) &= \kappa^{-1} \exp \sum_{A \subseteq J^*} \theta_A \prod_{(i,j) \in A} x_{ij} \\ &= \kappa^{-1} \exp \sum_{(i,j)} \theta_{ij} x_{ij} = \kappa^{-1} \exp \theta_L L(x), \end{aligned}$$

where $L(x) = \sum_{i < j} x_{ij}$ is the number of edges in x . Having imposed the homogeneity assumption for the case where $n = 20$ vertices, instead of the previous 190 configurations, we only need to consider counts of the edges. Once we impose homogeneity in this way (irrespective of the particular dependence assumption used), the statistics, $z_A(x)$, in Equation (7.1) become more than indicator variables for individual configurations. Rather, they rather reduce to counts in x of configurations of particular types, and then Equation (7.1) takes on the familiar form of an ERGM presented in Chapter 6.

For the Bernoulli model, the number of edges is *sufficient* information for the parameter θ_L , and hence sufficient information about the model defined by the Bernoulli dependence graph and the interactions. We say that the subgraph is a “sufficient subgraph” because its count is a sufficient

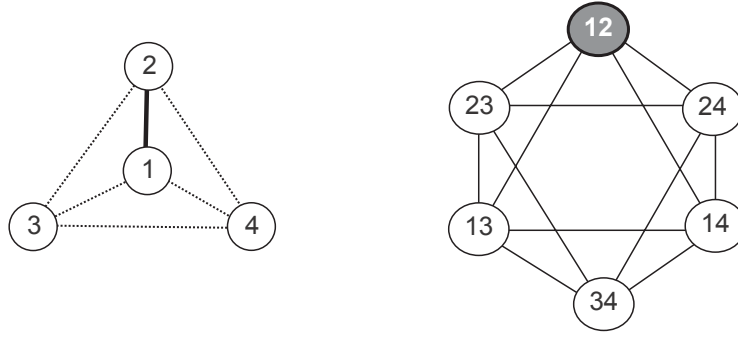


Figure 7.4. Singleton clique in dependence graph and corresponding configuration in X .

statistic in the same way that the number of heads in n flips of a (biased) coin is a sufficient statistic for the probability of heads – letting “1” denote “heads,” we do not need to know if the sequence was $(1,0,0,1,0)$ or $(0,1,0,0,1)$, only that the total was 2. Hence, the probability of observing a particular graph is dependent on the graph statistic $z(x)$ for x and corresponding parameter θ .

Markov Model. For the Markov model, just as in the Bernoulli model, the singleton cliques correspond to the individual tie-variables, such as 12 in Figure 7.4.

As described previously, the clique 12, 23 in D corresponds to X_{12} and X_{23} , a configuration in X that is a 2-star centered on vertex 2 whenever $x_{12} = x_{23} = 1$. In contrast, the subgraph in D that is induced by 12 and 34 is not complete because these two vertices are not tied. Cliques of size 3 in D correspond to two different configurations in X . For example, the 3-clique in Figure 7.5, consisting of nodes 12, 23, and 24, all have node 2 from G in common, and therefore correspond to a 3-star centered on node 2 in G . The clique 12, 23, 13 in Figure 7.6 corresponds to a triangle configuration in X .

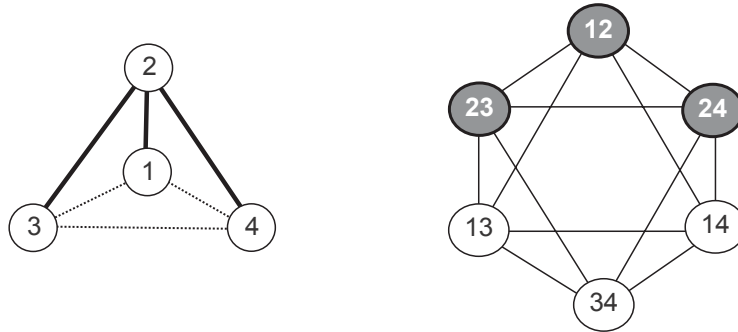


Figure 7.5. Three-clique in dependence graph and corresponding 3-star configuration in X .

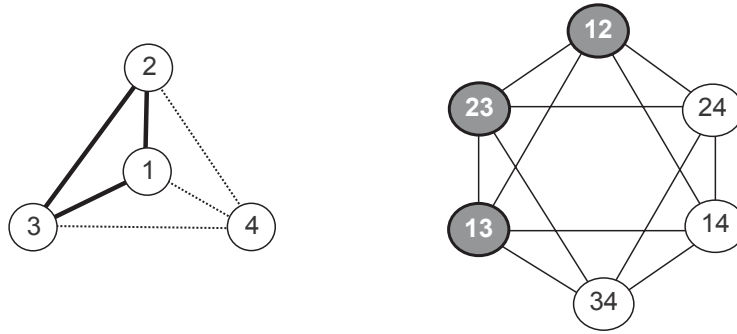


Figure 7.6. Three-clique in dependence graph and corresponding triangle configuration in X .

For Markov graph models, the cliques in D correspond to edges, stars, or triangles (Frank & Strauss, 1986). The Hammersley-Clifford theorem informs us that Markov random graphs are completely characterized by their edges, stars, and triangles, the sufficient subgraphs of the Markov random graph model. In other words, for any graph in X , its probability, under the Markov assumption, is completely determined by the numbers of edges, stars, and triangles. Whether there is, for example, a complete subgraph with five vertices is not important other than the fact that it contributes ten triangles, ten edges, and so on.

We reiterate that star and triangle configurations correspond to interactions of tie-variables that arise from the Markov dependence assumption. In each case, the interaction term informs us how much more likely (or unlikely) the particular configuration is in comparison to that expected from the probabilities of the respective tie-variables alone. As a result, in a properly hierarchical model, where higher-order interaction effects are included with appropriate lower-order effects, an interaction term gives us information about the presence of a configuration, taking into account the lower-order effects. There is a direct comparison here with a standard log-linear model. So, a 2-star parameter represents a centralization effect, taking into account the lower-order edge effect. In other words, inferences based on a 2-star parameter are conditional on the number of edges, so we can infer whether there are more than expected 2-stars given the number of edges (i.e., more than expected centralization given the density). Similarly, a 3-star parameter permits inferences, taking into account the numbers of 2-stars and edges. Importantly, a triangle parameter represents triangulation conditional on the number of 2-stars and edges. This permits inferences about network closure, given the number of density and the number of 2-paths (i.e., the potential for closure).

This is a remarkable result: given only a simple assumption regarding dependence between the ties, we do not need to know the exact structure of a network in order to know how likely it is, we only need to know some

of the subgraphs (edges, stars, triangles) of this network. Let us consider what listing the stars and triangles entail. For each of the n nodes, the other $n - 1$ nodes may have edges that can be combined in $\binom{n-1}{2}$ ways such that the node is the center of a 2-star, and hence there are $n\binom{n-1}{2}$ distinct possible 2-stars. In general, there are $n\binom{n-1}{k}$ distinct possible k -stars, for $k = 2, \dots, n - 1$. Because a triangle includes three nodes, the number of distinct possible triangles is given by $\binom{n}{3}$. If we were to allow for a separate parameter for each configuration corresponding to a clique in the dependence graph, we would have too many parameters. For example, a Markov graph on twenty vertices would have $\binom{20-1}{2} + 20\binom{20-1}{3} + 20\binom{20-1}{4} + \dots + 20\binom{20-1}{19} + \binom{20}{3} = 10,486,690$ distinct parameters but only 190 tie-variables.

As before, to reduce the number of distinct configurations and corresponding parameters, we impose homogeneity whereby all structurally identical graphs are equally probable. This means, for instance, that the effect for every 2-star is identical, so that there is one 2-star parameter, with a statistic that is the count of the number of 2-stars. Suppose we apply a Markov dependence assumption to Equation (7.1). Among the possible configurations A are 2-stars with terms $\sum_i \sum_{j < b, j, b \neq i} \theta_{ijb} x_{ij} x_{ib}$ in the equation. We impose homogeneity by letting $\theta_A = \theta_{S(2)}$ for all A corresponding to 2-stars. Then the summation across all 2-star terms reduces to $\theta_{S(2)} S_2(x)$, where $S_2(x)$ is the number of 2-stars in x . Similar results follow for the other configurations (edges, other stars and triangles), so that in the end Equation (7.1) reduces to the Markov model familiar from Chapter 6. Having imposed the homogeneity assumption, with twenty nodes, instead of 10 million subgraphs, we only need to consider counts of edges, of eighteen different star configurations and of triangle configurations. The numbers of edges, of stars of different orders, and of triangles are *sufficient* information for the parameters and hence constitute sufficient statistics for the model. It is worth emphasizing that imposing homogeneity allows us to consider graph topology (the structure of the graph) irrespective of the labeling of the graph nodes.

The homogeneity assumption reduces the model to n parameters, comprising an edge and triangle parameter and $(n - 2)$ star parameters. In most instances, we cannot have unique parameters for all stars. For most data, such a model will not be estimable (e.g., if no node is connected to every other node, the count of $(n - 1)$ -stars is zero and the parameter cannot be estimated). In any event, for a large network, n parameters is still not very parsimonious. One way of reducing the number of star parameters is by setting some of them to zero. Setting higher-order parameters to zero is analogous to ignoring higher-order interactions among predictors in a standard logistic regression. For example, Frank and Strauss (1986) suggested (as one possible model) the “triad model” with only an edge, a

2-star, and a triangle parameter. All 3-star and higher-order star parameters are set to zero. Robins, Pattison, and Woolcock (2005) suggested an extension that included a nonzero 3-star parameter. However, for many data sets, these models still cannot be estimated properly, as explained in Chapter 6. An alternative to restricting the number of star parameters is to constrain them into a single alternating star parameter. This is now the preferred way to proceed and helps, although does not eliminate, the problems with estimation. The details are set out in Chapter 6.

We have gone through Bernoulli and Markov dependence at some length because these simpler models illustrate the arguments well. However, we emphasize that for the practical purposes of fitting data, these models will only rarely be used with success. Social circuit models with alternating parameter constraints are by far the preferred option to avoid estimation difficulties.

Social Circuit Dependence. We have already mentioned how the social circuit partial dependence assumption was formalized for the case of random graph models by Pattison and Robins (2002) using partial dependence graphs. Snijders et al. (2006) proposed a model that satisfies the social circuit independence assumption. As described in connection with Figure 7.3, this model means that some tie-variables may be conditionally dependent given the presence of other ties, even if they do not share a node. As explained in Chapter 6, the model consists of all Markov parameters with the addition of parameters for k -triangles and k -independent 2-paths. Other configurations are also consistent with the dependence assumptions but to date have not been widely used (e.g., k -cliques).

Analogous to the case of Markov graphs, allowing for configurations that correspond to all distinct k -triangles would lead to too many statistics, and hence a homogeneity restriction is imposed so that all structurally identical graphs have the same probability under the model. Still, with this homogeneity restriction, the number of triangle parameters will be too large. Using the same rationale as for the introduction of alternating stars, a restriction can be imposed on the k -triangle parameters to produce the alternating triangle parameter, as explained in Chapter 6.

Sufficient Subgraphs for Directed Graphs. Using the same approach as described previously, we can derive sufficient statistics for directed graphs. The important thing to bear in mind is that there are twice as many tie-variables for a directed graph, and thus, there are twice as many nodes in the dependence graph (Figure 7.7). Because there are now two tie-variables for each node pair, there is a greater variety of different star and triangle configurations in Markov dependence. The singleton cliques correspond to a (directed) arc, and cliques 12 and 21 in Figure 7.7 correspond to two distinct configurations.

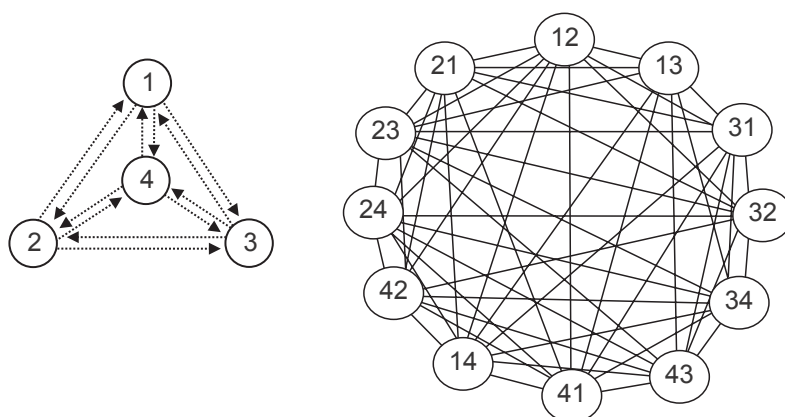


Figure 7.7. Tie-variables for directed graph on four vertices with corresponding Markov dependence graph.

Let us return to the example of a four-node graph using a Markov dependence assumption. Applying the homogeneity restriction as discussed previously, the sufficient subgraphs derived from D in Figure 7.7 are given in Figure 7.8.

In addition to arcs, there is another dyadic configuration, namely, reciprocated arcs (cliques in D of the type $\{12, 21\}$). There are also various star configurations, as well as seven triangle statistics representing different types of relational closure in a subgroup of three actors. As can be seen, parameters for these configurations essentially parameterize the triad census of a graph (Frank, 1979; Frank & Strauss, 1986; Holland & Leinhardt, 1981; Wasserman & Faust, 1994). If the parameters corresponding to configurations with more than three vertices are set to zero, the directed Markov graph model may be parameterized directly in terms of the triad census.

The social circuit dependence assumption for directed graphs builds naturally on that of the partial dependence approach described previously in this chapter. We do not go into details here. Suffice it to say that the sufficient subgraphs include the directed k -triangle and k -path configurations described in Chapter 6. The alternating or geometrically weighted form of the parameters is the most commonly used in fitting data.

7.3 Dependence Graphs Involving Attributes

Attribute variables can also be introduced into dependence graphs, either as exogenous predictors of network ties (social selection effects) or outcome variables from network structure (social influence effects). These types of models are presented in Chapters 8 and 9. There are some additional complexities in introducing exogenous predictors and different variable types into a dependence graph formulation that are beyond the

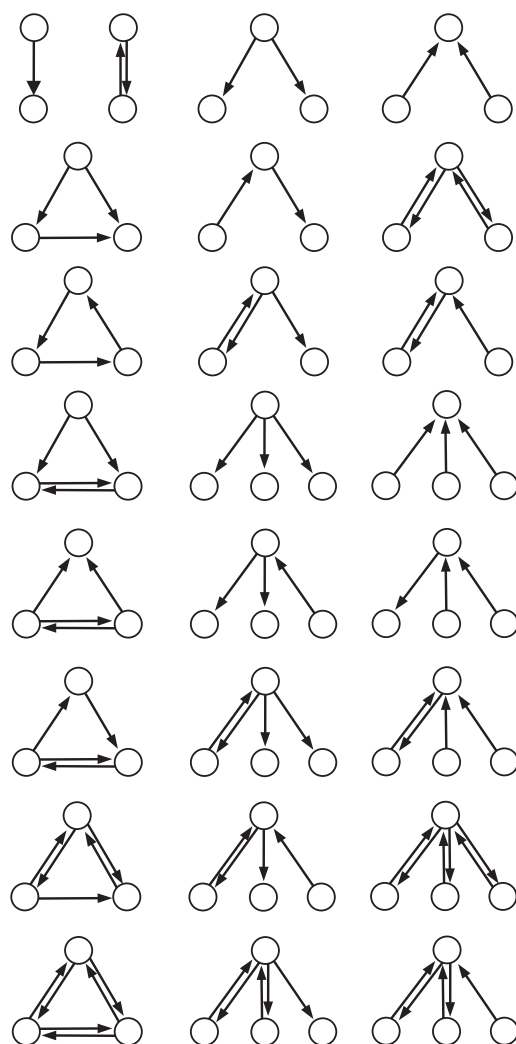


Figure 7.8. Sufficient subgraphs for directed Markov graph on four vertices.

scope of this chapter. Interested readers may consult Robins, Pattison, and Elliott (2001); Robins, Elliott, and Pattison (2001); and Robins and Pattison (2005).

7.4 Conclusion

The ERGM is a stochastic model in which the observed network is regarded as one realization from a probability distribution on the set of possible networks or graphs $\mathbf{X}$ on a fixed set of nodes. The probability of any particular realization x depends on parameters and statistics associated with certain configurations of x . These configurations are not

arbitrarily chosen; on the contrary, they are seen as the outcomes of particular social processes that give rise to the network.

The process of formulating ERGMs for networks may be seen as comprising the following steps. First, a dependence structure is postulated in the form of dependence graph. Second, cliques are derived from the dependence graphs. Third, the Hammersley-Clifford theorem is applied to provide a factorization with parameters based on cliques. Fourth, some homogeneity constraints are proposed to identify the model. As a result, the general form of the model has parameters and statistics relating to the presence of various types of local network configurations.

Elaborations of this basic form of the model permit different forms of dependence assumptions among network variables to be incorporated in a hierarchy of model forms. In addition, exogenous variables (e.g., node attributes or geographic location) can be built in. This general approach to model building also permits the construction of models for patterns of social influence within a network (i.e., models that allow individual-level attributes to be predicted from network ties).

Work on other possible forms of dependence is ongoing. The advantages of a systematic approach of understanding dependence are more than theoretic: the type of data that need to be collected in a snowball sample to estimate a large-scale ERGM is informed by the dependence structure. Moreover, it is possible to identify a hierarchy of dependence assumptions and so investigate further model elaborations systematically. These issues are taken up in the final chapter of this book.